# MECH 325
# Flexible Drive Elements



# Objectives

By the end of this section, you should be able to:

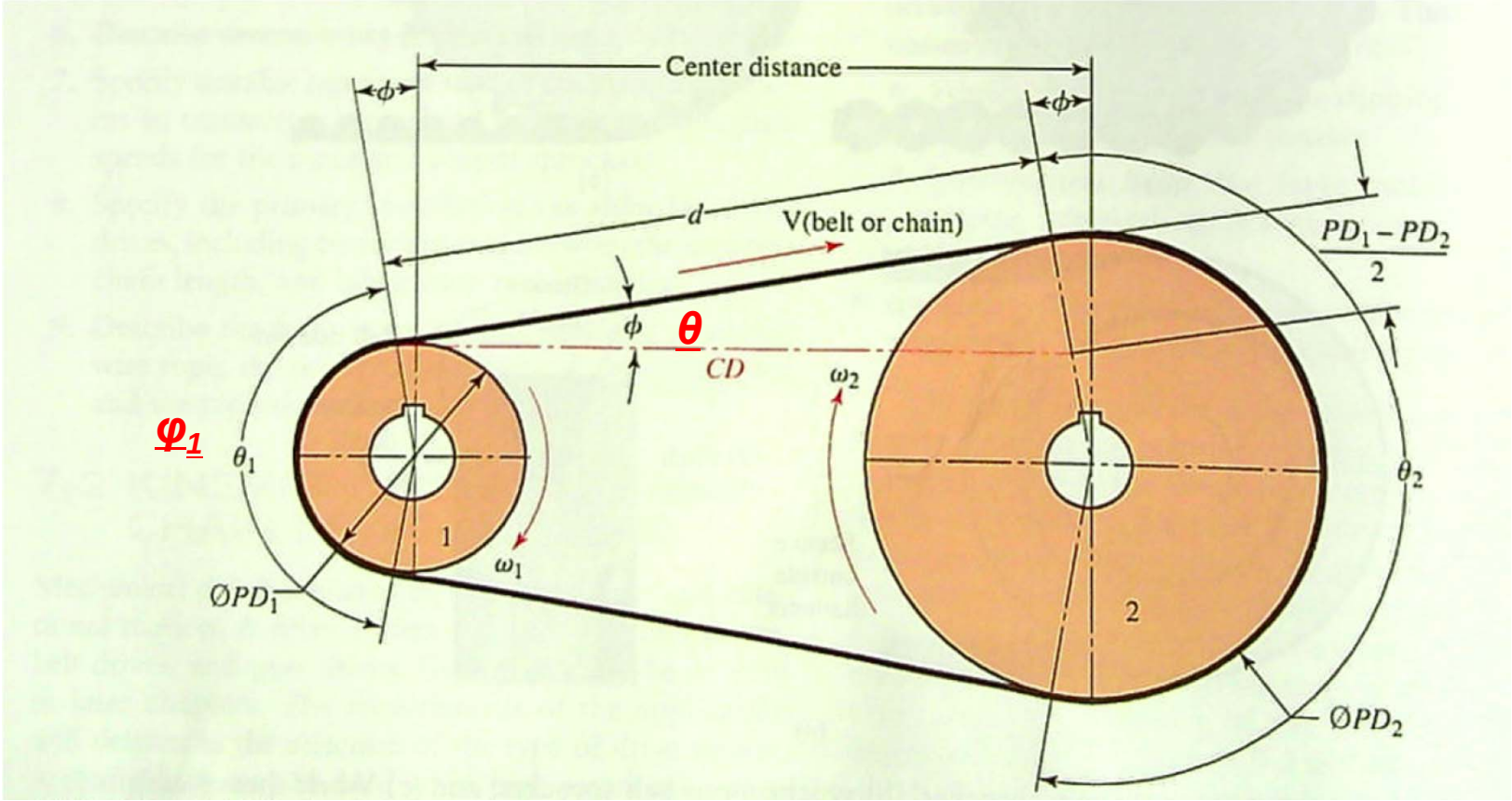
- Describe the characteristics and uses of:
  - Flat belts
  - V-belts
  - Timing belts
  - Roller chains
- Analyze flexible drives in terms of power, speed, forces, and life
- Select and size an appropriate flexible drive for a given application

# Flat Belts



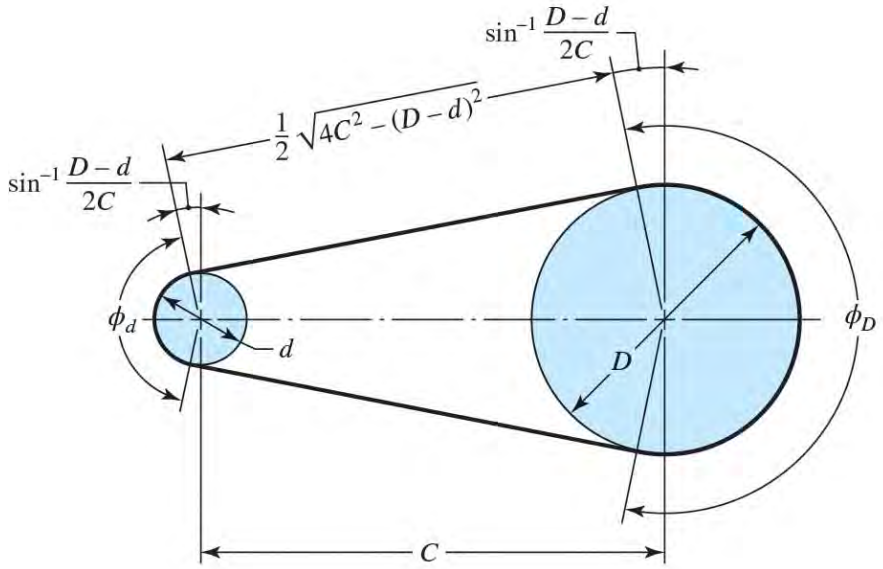
- Flat belts can be used for power transmission.
- Flat belts are less popular compared to V-belts, timing belts and chains
- Flat belt are extensively used in material handling

# Belt Kinematics




PD = Pitch Diameter  Phi (theta) = Wrap Angle  V = Belt(Chain) Speed  Theta (phi) = Belt Angle

# Flat-Belt Geometry – Open Belt



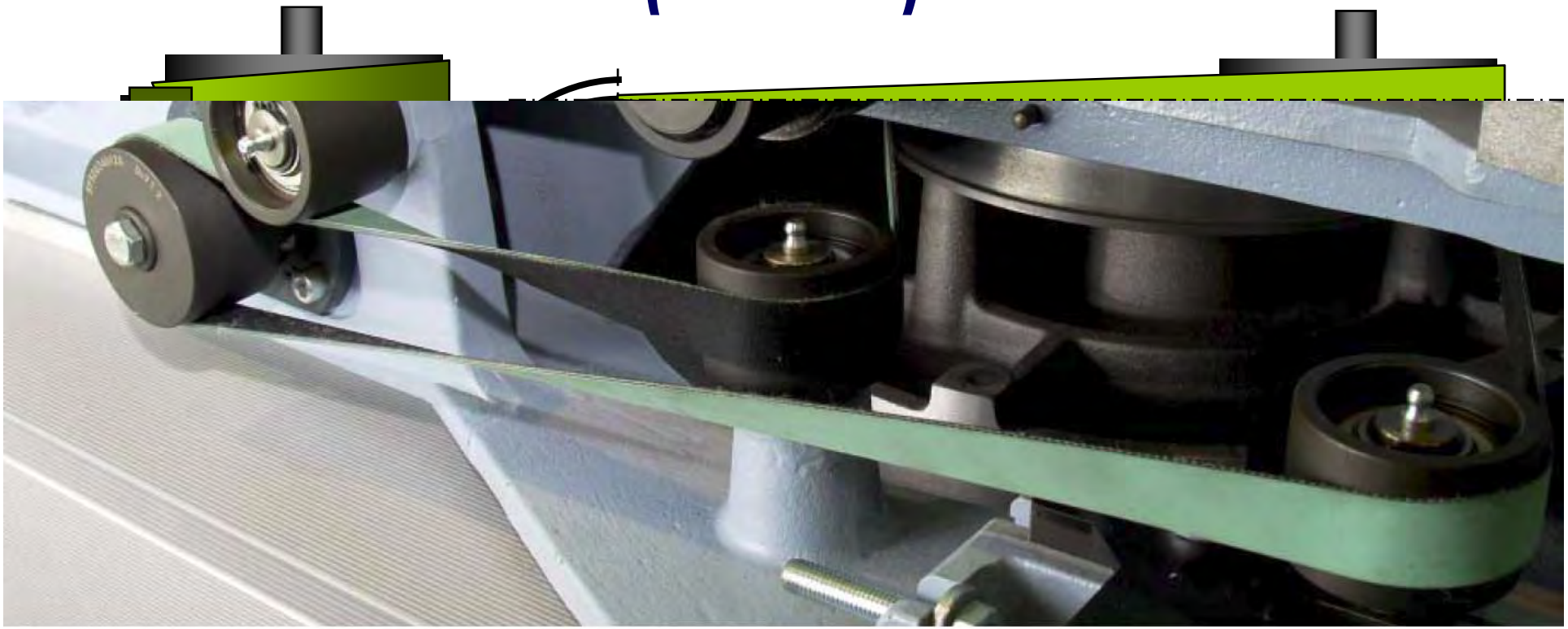

(a)

$$\phi_d = \pi - 2\sin^{-1}\frac{D-d}{2C}$$

$$\phi_D = \pi + 2\sin^{-1}\frac{D-d}{2C} \qquad \textbf{(17 - 1)}$$

$$L = \left[4C^2 - (D-d)^2\right]^{1/2} + \frac{1}{2}(D\phi_D + d\phi_d) \qquad \textbf{(17 - 2)}$$

# Flexible Drive Configurations (cont'd)



Twisted flat belt drive on a textile machine

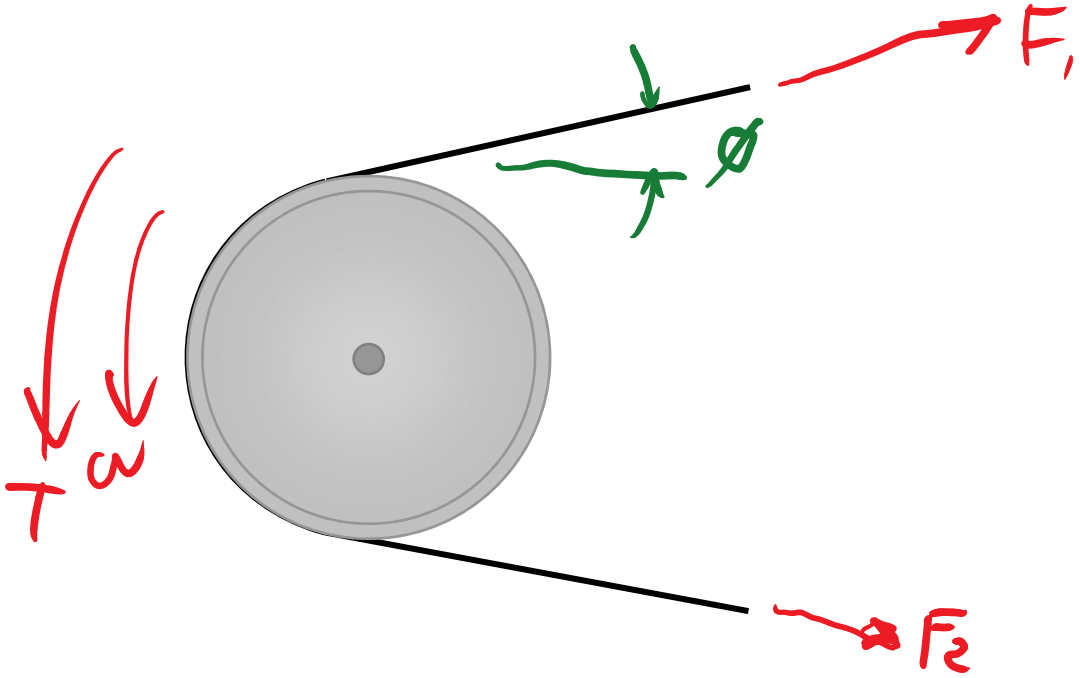


**Driven (Keyed to Shaft)**

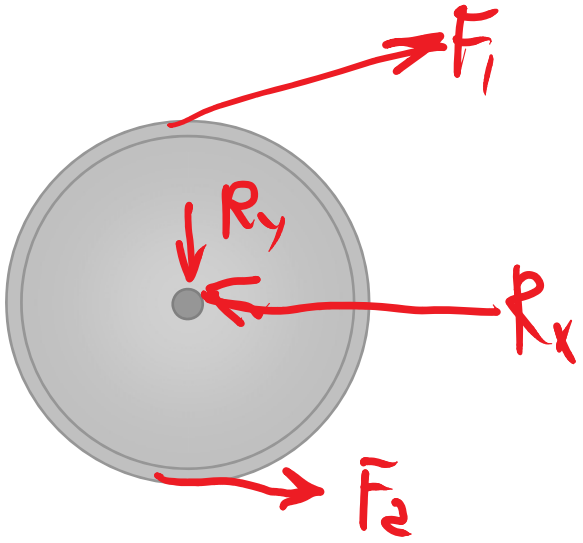
**Clutch-less disconnect**

# Belt Power Transmission

Transmission Parameters



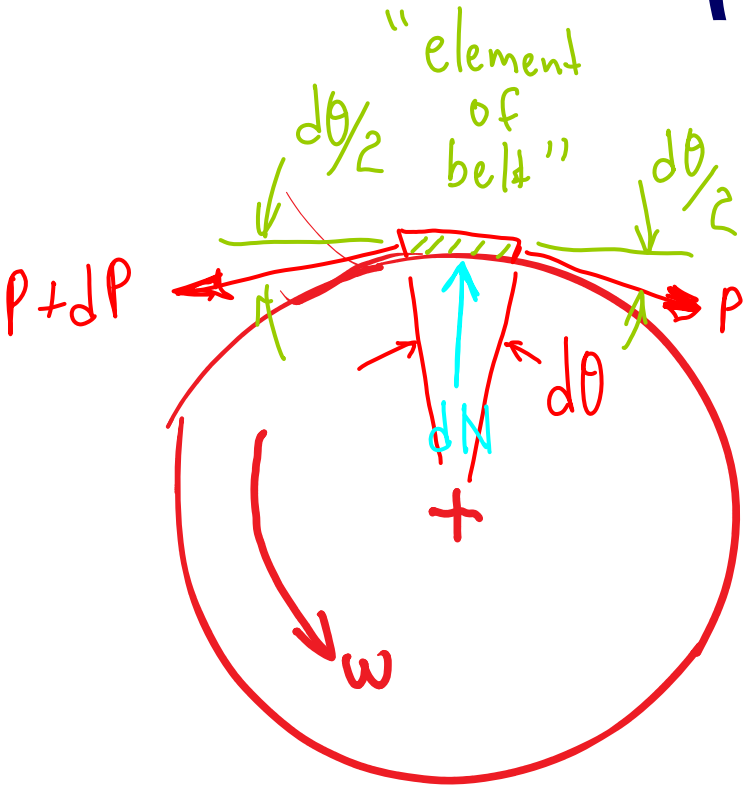

Pulley FBD




$R_x = (F_1 + F_2) \cos \phi$

$R_y = (F_1 - F_2) \sin \phi$

# Friction Belt Tension Derivation (Capstan Equation)




$$dN \approx P\sin\frac{d\theta}{2} + P\sin\frac{d\theta}{2}$$

$$dN = 2P\,\frac{d\theta}{2}$$

$$dN = P\,d\theta$$

- Recall $\sin\theta = \theta$

# Capstan Equation

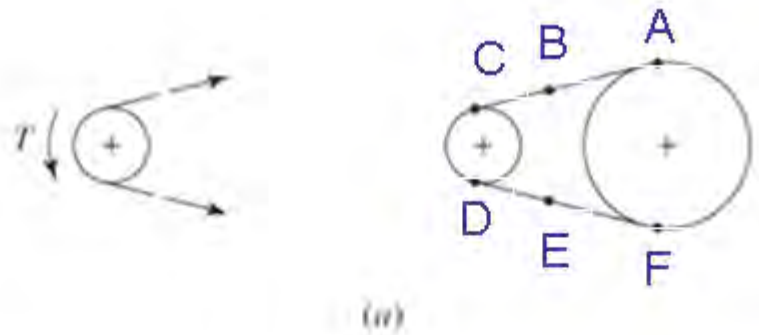


**Capstan Winch on a Sailboat**

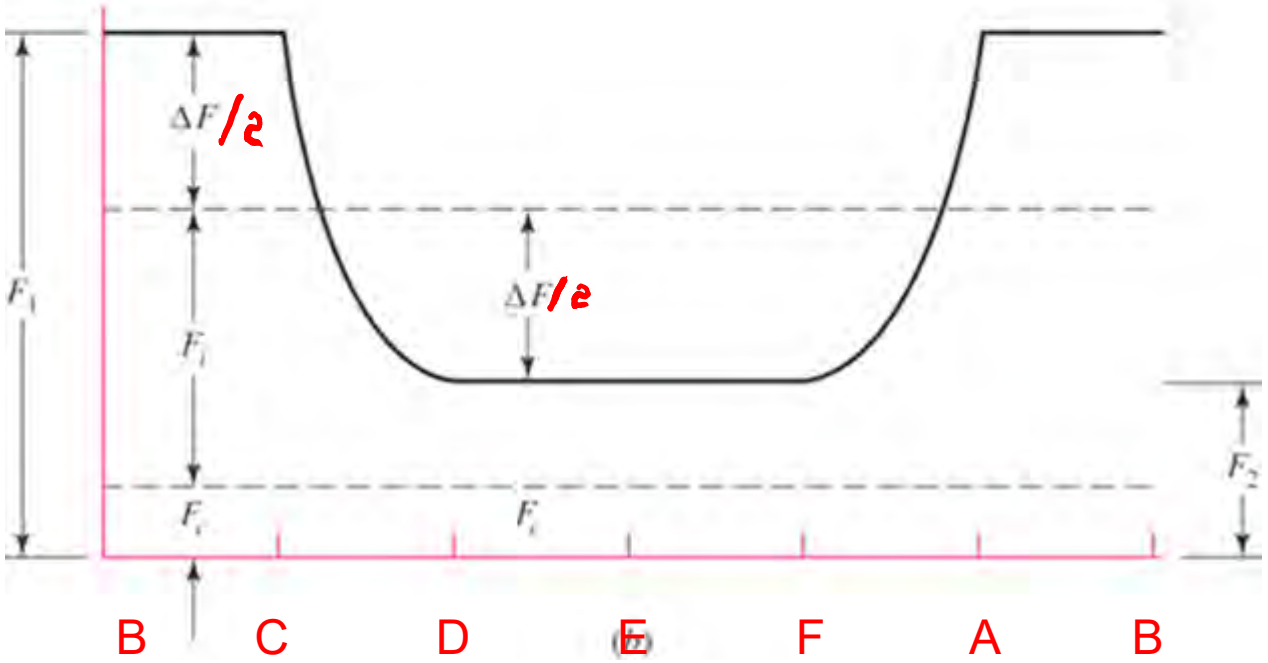


**Split Drum for Mooring Ships**

# Flat Belt Tensile Forces



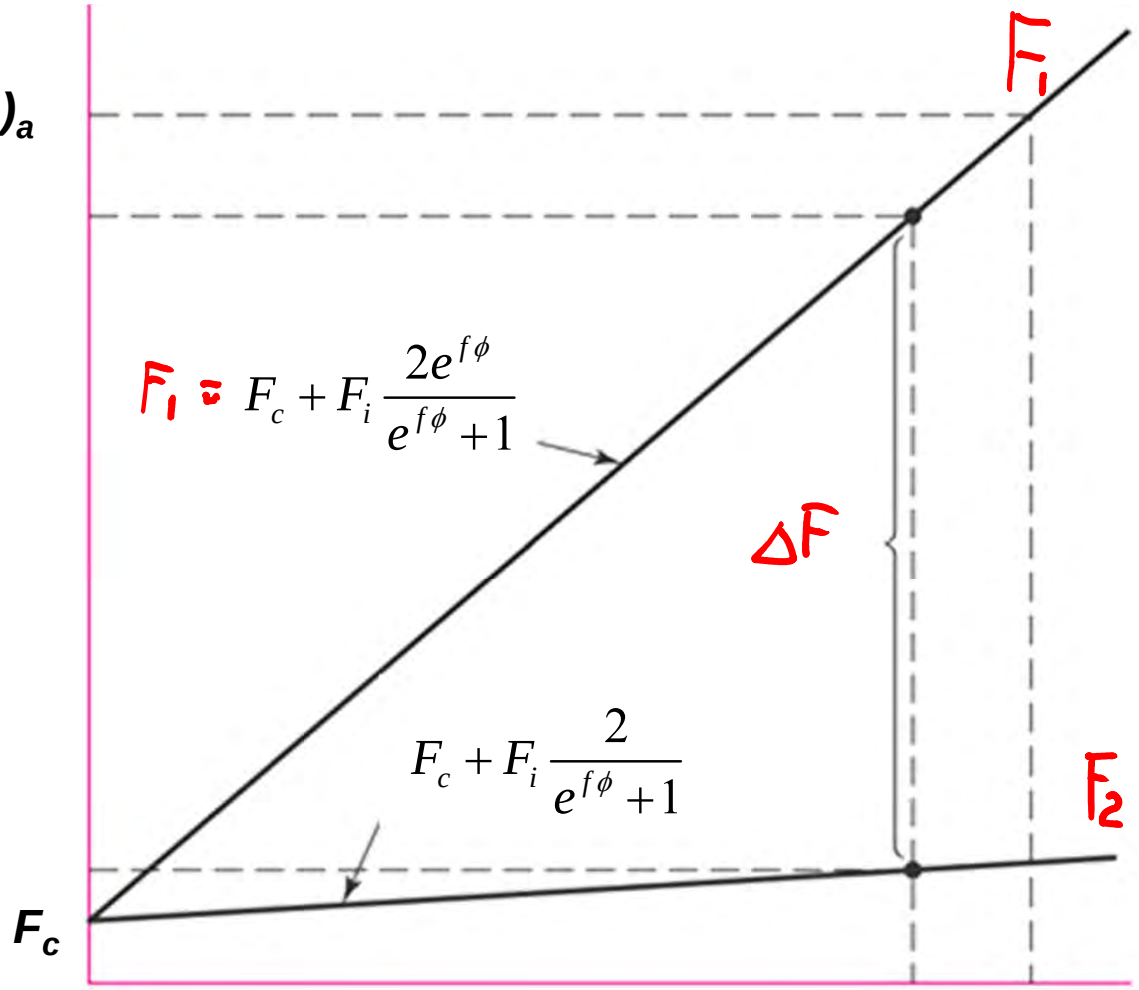

**Identify** A-F on the chart




**Identify:**

Slack-side tension
Taut-side tension
Centrifugal tension
Pretension

# Belt Tensions




$f$ = const.

$\omega$ = constant

# Flat Belt Equations

- Forces

$$\frac{F_1 - F_c}{F_2 - F_c} = e^{f\phi} \qquad F_c = \frac{w}{32.17}\left(\frac{V}{60}\right)^2$$

- Torque

$$T = \frac{d}{2}\left(F_1 - F_2\right) \qquad T = F_i d\,\frac{e^{f\phi}+1}{e^{f\phi}-1}$$

- Power

$$H = \frac{\left(F_1 - F_2\right)V}{33{,}000} \qquad T\,(in-lbs) = 63025\,\frac{H\,(HP)}{n\,(rpm)}$$

# Flat Belt Design (Shigley)

$(F_1)_a = bF_aC_pC_v$

| | |
|---|---|
| $(F_1)_a$ | largest allowable tension |
| $b$ | belt width |
| $F_a$ | allowable tension/unit width (manufacturers ratings) |
| $C_p$ | pulley correction factor (Tab. 17-4) |
| $C_v$ | velocity correction (Fig. 17-9) (leather belts only) |

$H_a = H_{nom}K_sn_d$

| | |
|---|---|
| $H_{nom}$ | nominal (rated) power |
| $H_a$ | design power |
| $K_s$ | service factor (Table 17-15) |
| $n_d$ | design ~~safety~~ factor |

Want $F_1 < (F_1)_a$

# Flat Belt Design (Shigley)

**Pulley Correction Factor $C_P$ for Flat Belts**

| | Small-Pulley Diameter, in | | | | | |
|---|---|---|---|---|---|---|
| **Material** | **1.6 to 4** | **4.5 to 8** | **9 to 12.5** | **14, 16** | **18 to 31.5** | **Over 31.5** |
| Leather | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 | 1.0 |
| Polyamide, F–0 | 0.95 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 |
| F–1 | 0.70 | 0.92 | 0.95 | 1.0 | 1.0 | 1.0 |
| F–2 | 0.73 | 0.86 | 0.96 | 1.0 | 1.0 | 1.0 |
| A–2 | 0.73 | 0.86 | 0.96 | 1.0 | 1.0 | 1.0 |
| A–3 | – | 0.70 | 0.87 | 0.94 | 0.96 | 1.0 |
| A–4 | – | – | 0.71 | 0.80 | 0.85 | 0.92 |
| A–5 | – | – | – | 0.72 | 0.77 | 0.91 |

*Source:* Average values of $C_P$ for the given ranges were approximated from curves in the *Habasit Engineering Manual*, Habasit Belting, Inc., Chamblee (Atlanta), Ga.

Table 17–4

**Suggested Service Factors for V-Belt Drives**

| **Driven Machinery** | **Source of Power Normal Torque Characteristic** | **Source of Power High or Nonuniform Torque** |
|---|---|---|
| Uniform | 1.0 to 1.2 | 1.1 to 1.3 |
| Light shock | 1.1 to 1.3 | 1.2 to 1.4 |
| Medium shock | 1.2 to 1.4 | 1.4 to 1.6 |
| Heavy shock | 1.3 to 1.5 | 1.5 to 1.8 |

Table 17–15

Use this Table for Flat Belts

# Flat Belt Design (Shigley)

$(F_1)_a$

| Material | Specification | Size, in | Minimum Pulley Diameter in | Allowable Tension per Unit Width at 600 ft/min, lbf/in | Specific Weight, lbf/in$^3$ | Coefficient of Friction |
|---|---|---|---|---|---|---|
| Leather | 1 ply | $t = \frac{11}{64}$ | 3 | 30 | 0.035 to 0.045 | 0.4 |
| | | $t = \frac{13}{64}$ | $3\frac{1}{2}$ | 33 | 0.035 to 0.045 | 0.4 |
| | 2 ply | $t = \frac{18}{64}$ | $4\frac{1}{2}$ | 41 | 0.035 to 0.045 | 0.4 |
| | | | | 50 | 0.035 to 0.045 | 0.4 |
| | | | | 60 | 0.035 to 0.045 | 0.4 |
| | | $t = \frac{20}{64}$ | 6[a] | | | |
| | | $t = \frac{23}{64}$ | 9[a] | | | |
| Polyamide[b] | F–0[c] | $t = 0.03$ | 0.60 | 10 | 0.035 | 0.5 |
| | F–1[c] | $t = 0.05$ | 1.0 | 35 | 0.035 | 0.5 |
| | F–2[c] | $t = 0.07$ | 2.4 | 60 | 0.051 | 0.5 |
| | A–2[c] | $t = 0.11$ | 2.4 | 60 | 0.037 | 0.8 |
| | A–3[c] | $t = 0.13$ | 4.3 | 100 | 0.042 | 0.8 |
| | A–4[c] | $t = 0.20$ | 9.5 | 175 | 0.039 | 0.8 |
| | A–5[c] | $t = 0.25$ | 13.5 | 275 | 0.039 | 0.8 |
| Urethane[d] | $w = 0.50$ in | $t = 0.062$ | See Table 17–3 | 5.2[e] | 0.038 to 0.045 | 0.7 |
| | $w = 0.75$ in | $t = 0.078$ | | 9.8[e] | 0.038 to 0.045 | 0.7 |
| | $w = 1.25$ in | $t = 0.090$ | | 18.9[e] | 0.038 to 0.045 | 0.7 |
| | Round | $d = \frac{1}{4}$ | See Table 17–3 | 8.3[e] | 0.038 to 0.045 | 0.7 |
| | | $d = \frac{3}{8}$ | | 18.6[e] | 0.038 to 0.045 | 0.7 |
| | | $d = \frac{1}{2}$ | | 33.0[e] | 0.038 to 0.045 | 0.7 |
| | | $d = \frac{3}{4}$ | | 74.3[e] | 0.038 to 0.045 | 0.7 |

# Friction Development

- $f$ = coefficient of friction (0.4 to 0.8)
  - From belt and pulley materials (plus age and condition)
- $f'$ = "friction development"
- $f'$ = equivalent value of $f$ for which slipping occurs (calculate after an actual belt has been selected to verify design.

$$f' = \frac{1}{\phi} \ln \frac{F_1 - F_c}{F_2 - F_c}$$

note: symbol $f'$ also used for V-belt "effective friction"

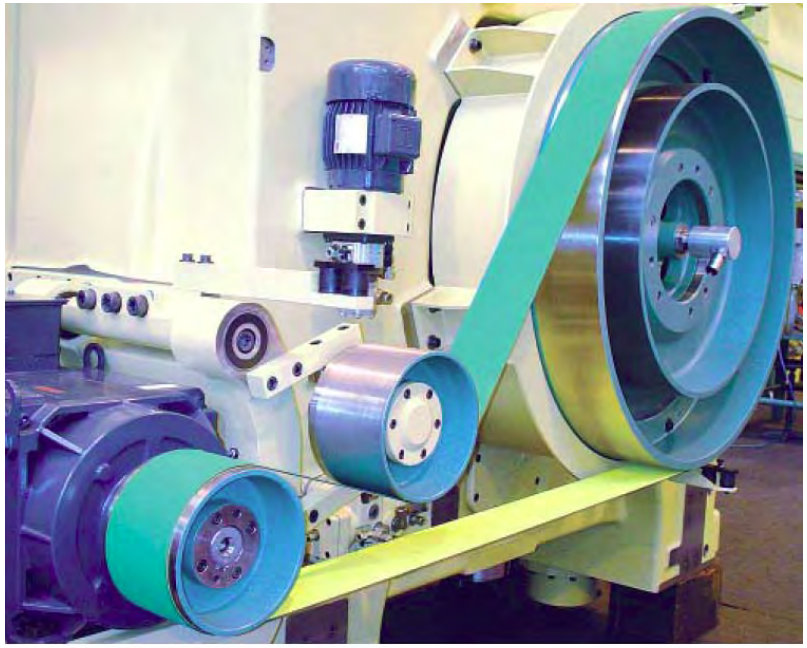
If $f > f'$ we can expect no slipping to occur

# Ref: Flat Belt Analysis Procedure

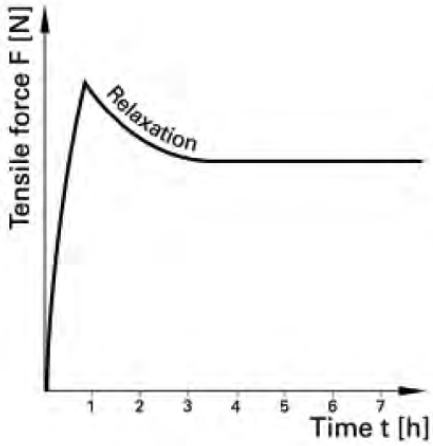
1. Find $e^{f\varphi}$ (from belt-drive geometry and friction)
2. Find $F_c$ (from belt-drive geometry and speed)
3. Find T (from $63{,}025H_{nom}K_s n_d/n$)
4. Find $F_{1a} - F_2$ (from $2T/d$). Assume $F_1$ is max. ($F_{1a}$)
5. Find $F_{1a}$ (from $F_a b C_p C_v$)
6. Find $F_2$
7. Find $F_i$ (from $(F_1 + F_2)/2 - F_c$)
8. Check if $f > f'$
9. Find $n_{fs}$ (from $H_a/(H_{nom}K_s)$ (confirm $n_d$)

# Flat Belt (Habisit Co.)

- Flat Belts require crowned pulleys
- Belts have a spring constant, $K_{1\%a.r.i}$ (after run-in) to accommodate material creep



Flat belt driven 160 ton press




$k_{1\%\,a.r.i.}$ = tensile force for 1% elongation per unit of width after running-in [N/mm]

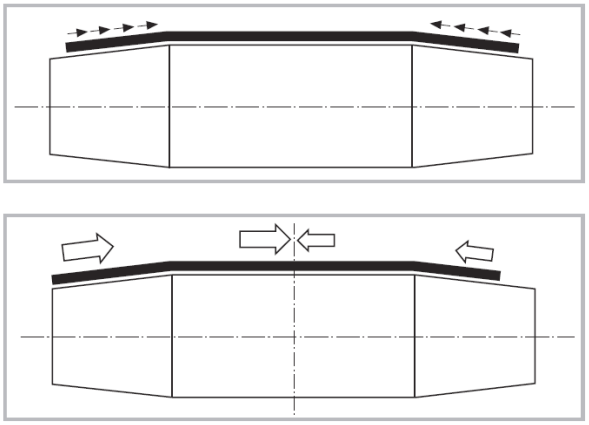
The $k_{1\%\,a.r.i.}$ is the decisive value for calculating the required belt width and the resulting shaft load after belt relaxation.

# Self-Tracking for Flat Belts

Recommended crown height:

Power transmission belts (rounded crowns)

| Pulley diameter acc. to ISO 22 | Recommended crown height (h) [mm] | | | | | Crown height acc. to ISO 22 **Not recommended** [mm] | |
|---|---|---|---|---|---|---|---|
| | Pulley width b acc. to ISO 22 [mm] | | | | | Pulley width [mm] | |
| [mm] | 20, 25, 32, 40, 50 | 63, 71, 80, 90, 100 | 112, 125, 140, 160 | 180, 200, 244, 250 | 280, 315, 355, 400 | b ≤ 250 | b ≥ 280 |
| 40 | 0.3 | 0.3 | 0.3 | | | 0.3 | 0.3 |
| 50 | | | | | | | |
| 63 | | | | | | | |
| 71 | | | | | | | |
| 80 | | | | | | | |
| 90 | | | | | | | |
| 100 | | | | 0.4 | | | |
| 112 | | | 0.4 | | | | |
| 125 | | 0.4 | | | | 0.4 | 0.4 |
| 140 | 0.4 | | | | 0.5 | | |
| 160 | | | | 0.5 | | 0.5 | 0.5 |
| 180 | | | 0.5 | | 0.6 | | |
| 200 | | | | | | 0.6 | 0.6 |
| 224 | | 0.5 | | 0.6 | | | |
| 250 | | | | | 0.7 | 0.8 | 0.8 |
| 280 | 0.5 | | | | | | |
| 315 | | | 0.6 | | | | |
| 355 | | | | | 0.8 | 1.0 | 1.0 |
| 400 | | 0.6 | | 0.7 | | | |
| 450 | | | | | | | |

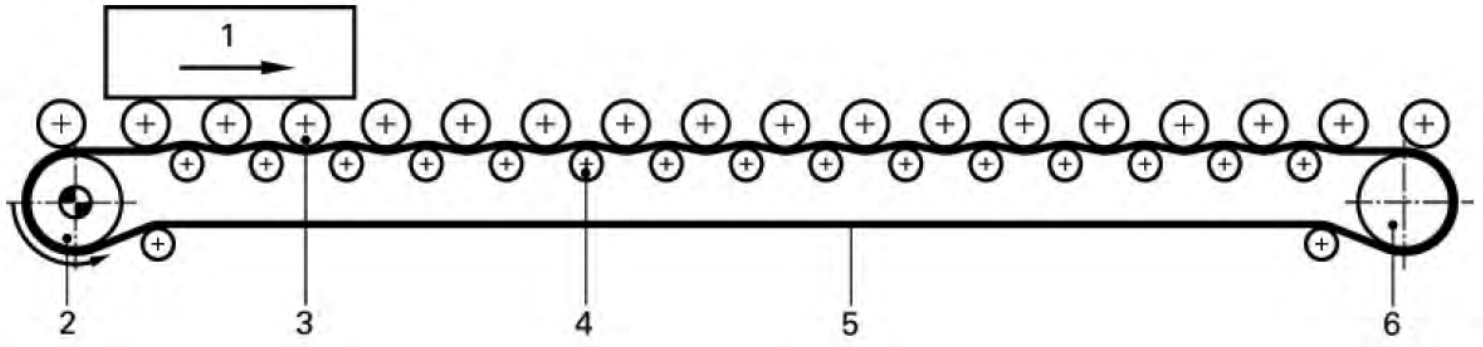


Trapezoidal Crowning

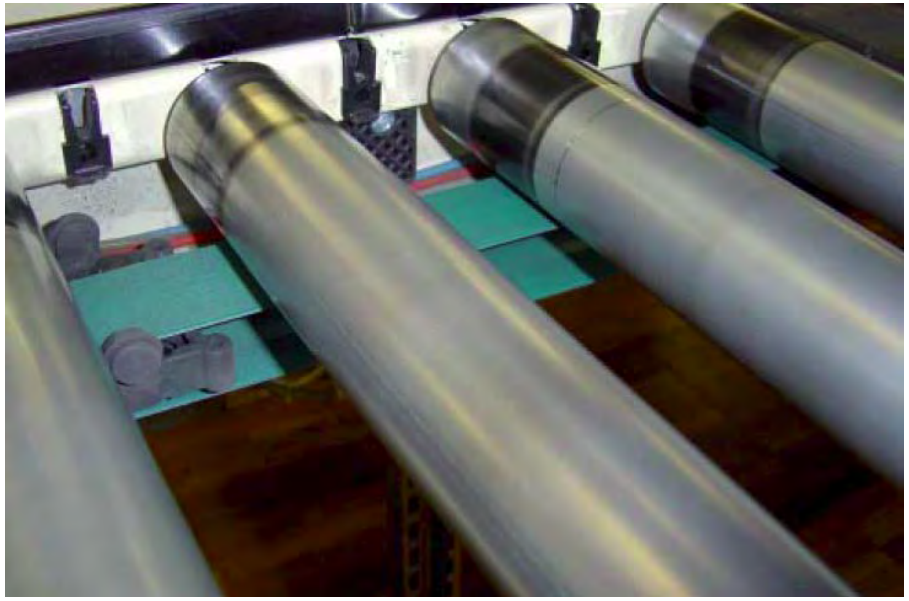
A movement of the belt to one side will create an imbalance of net forces that will move the belt back to the center (self-tracking).

Trapezoidal crowning is used primarily for Conveyor Belts.

# Live Roller Conveyor Drive




1 Transported piece good
2 Driving pulley
3 Carrying rollers
4 Pressure rollers
5 Driving belt
6 Return pulley



| | Belt group | Belt code | Double-sided power transmission, tangential drives | Thickness [mm] | Pulley diameter minimum with counter flection [mm] | Tensile force for 1% elongation per unit of width [N/mm] ($k_{1\%}$ after running in) | Nominal peripheral force per unit of width [N/mm] $(F_1)_a$ |
|---|---|---|---|---|---|---|---|
| Polyester power transmission belts | TC- | TC-10EF | ● | 1.8 | 25 | 5 | 10 |
| | | TC-20EF | ● | 2.0 | 25 | 10 | 21 |
| | | TC-20/25EF | ● | 2.5 | 50 | 11 | 23 |
| | | TC-35ER | ● | 2.5 | 50 | 18 | 38 |
| | | TC-35/30ER | ● | 3.0 | 50 | 18 | 38 |
| | | TC-35/35ER | ● | 3.5 | 70 | 18 | 38 |
| | | TC-55ER | ● | 3.0 | 70 | 25 | 53 |
| | TCF- | TCF-20EL | o | 2.6 | 80 | 10 | 21 |
| | | TCF-35EL | o | 3.0 | 100 | 18 | 38 |
| | | TCF-55EL | o | 3.5 | 150 | 25 | 53 |
| | | TCF-20H | ● | 1.5 | 25 | 10 | 20 |
| | | TCF-50H | ● | 2.0 | 60 | 24 | 48 |
| Polyamide power transmission belts **Nylon** | S- | S-10/15 | ● | 1.5 | 40 | 4.4 | 12 |
| | | S-18/20 | ● | 2.0 | 60 | 8 | 22 |
| | | S-18/30 | ● | 3.0 | 60 | 8 | 22 |
| | | S-33/40 | ● | 4.0 | 125 | 13 | 37 |
| | | S-33/50 | ● | 5.0 | 125 | 13 | 37 |
| | | S-140H | ● | 1.7 | 40 | 4.8 | 13 |
| | | S-141H | ● | 2.3 | 40 | 4.8 | 13 |
| | | S-250H | ● | 2.3 | 100 | 11 | 29 |
| | | S-250HR | ● | 2.6 | 100 | 11 | 29 |
| | | S-251H | ● | 3.0 | 100 | 11 | 29 |
| | | S-321H | ● | 3.2 | 125 | 13 | 35 |
| | | S-390H | ● | 3.2 | 160 | 14 | 38 |
| | | S-391H | ● | 4.0 | 160 | 14 | 38 |
| | A- | A-2 | o | 2.7 | 60 | 7.5 | 22 |
| | | A-3 | o | 3.4 | 125 | 12 | 36 |
| | | A-4 | o | 5.0 | 300 | 21 | 63 |
| | | A-5 | o | 6.8 | 450 | 30 | 92 |
| Aramide power transmission belts **Kevlar** | TF- | TF-10 | ● | 1.7 | 25 | 10 | 10 |
| | | TF-15 | ● | 2.0 | 30 | 15 | 15 |
| | | TF-15H | ● | 1.5 | 30 | 15 | 15 |
| | | TF-22 | ● | 2.4 | 60 | 22 | 22 |
| | | TF-33 | ● | 3.0 | 100 | 33 | 33 |
| | | TF-50 | ● | 3.9 | 125 | 50 | 50 |
| | | TF-75T TF-75TE | o | 4.4 | 200 | 75 | 75 |

● Applicable o Not applicable

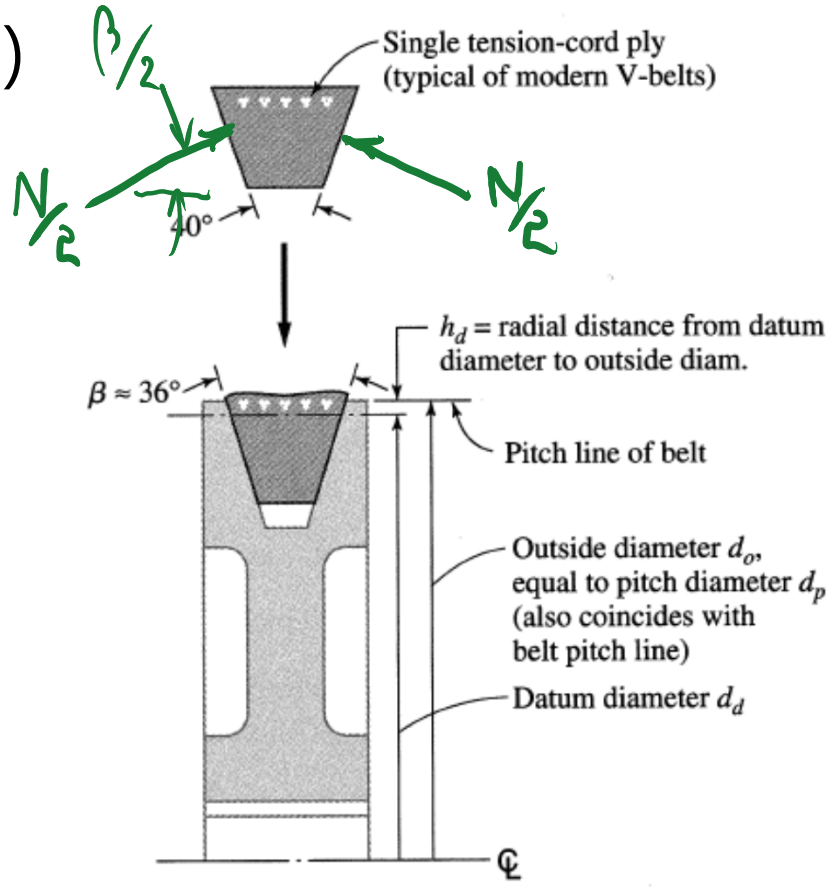
# V-Belt

- V-belts generate more friction than flat belts
- η ~70-94% (98% flat belts)
- Effective friction

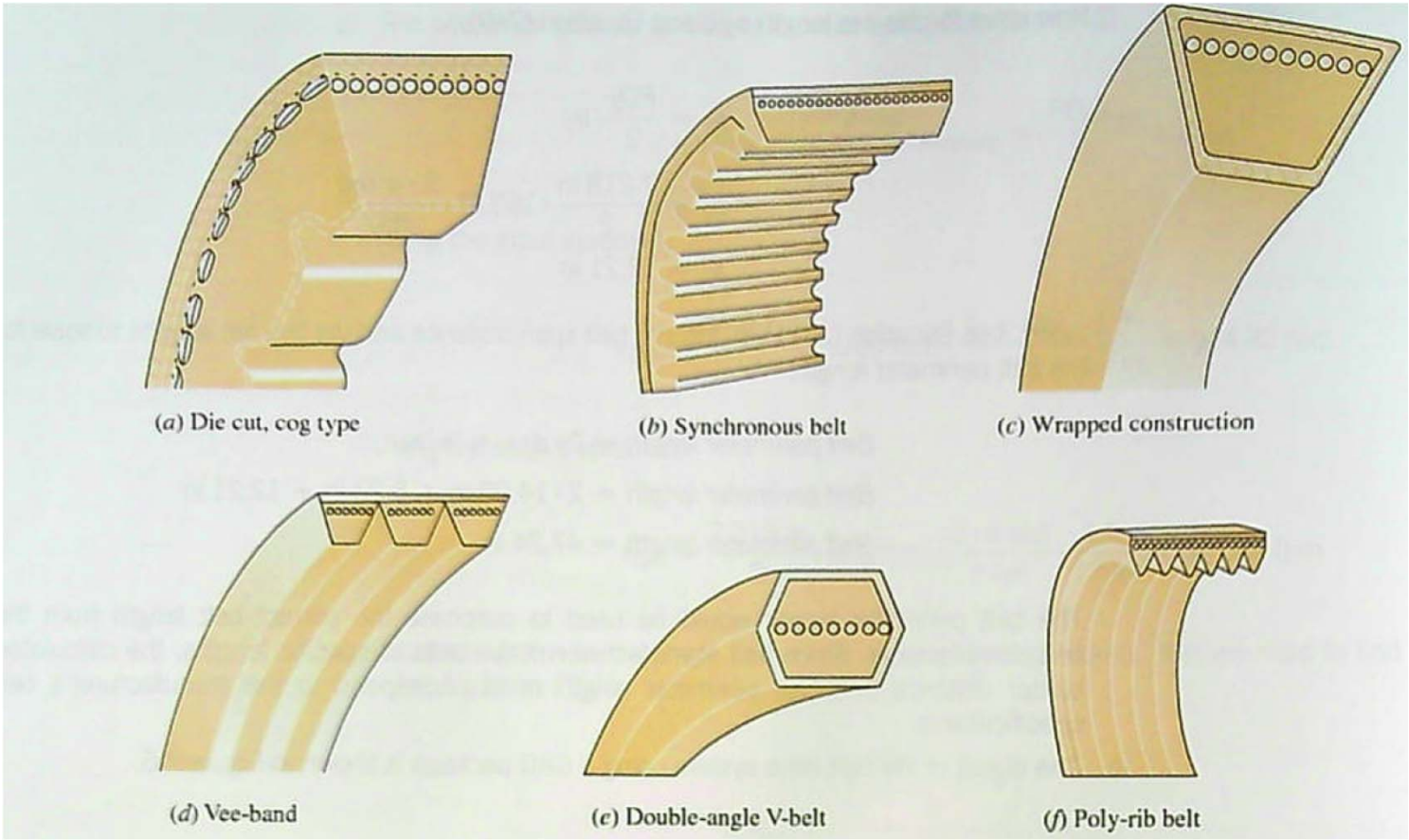
$$f' = \frac{f}{\sin(\beta/2)}$$

- The expressions from flat belts still apply:

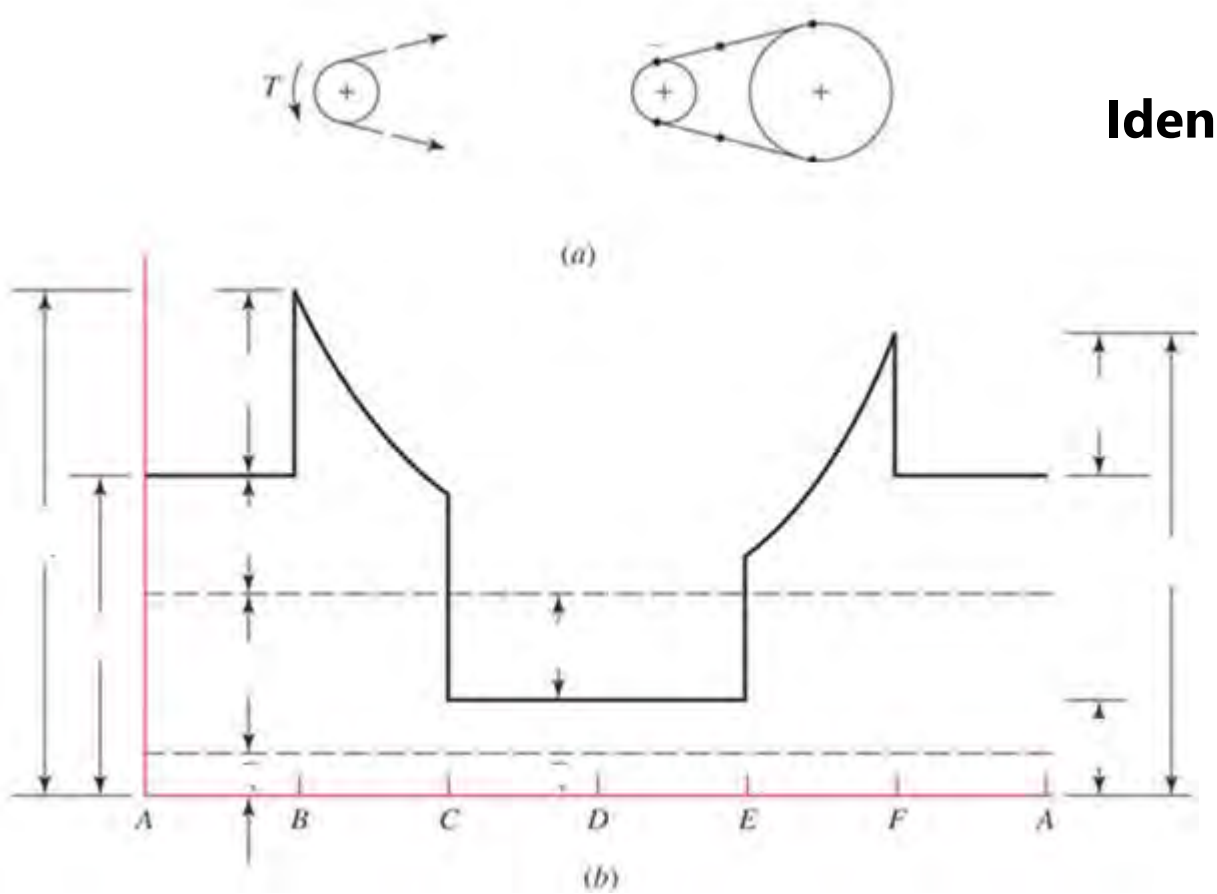
$$\frac{F_1 - F_c}{F_2 - F_c} = e^{f'\phi}$$



# Belt Styles


(a) Die cut, cog type
(b) Synchronous belt
(c) Wrapped construction
(d) Vee-band
(e) Double-angle V-belt
(f) Poly-rib belt

# V-Belt Tensile Forces

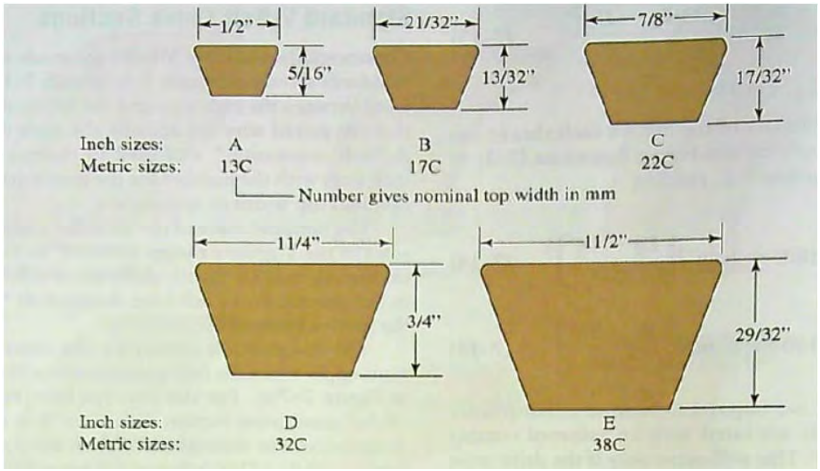



**Identify** A-F on the drawing

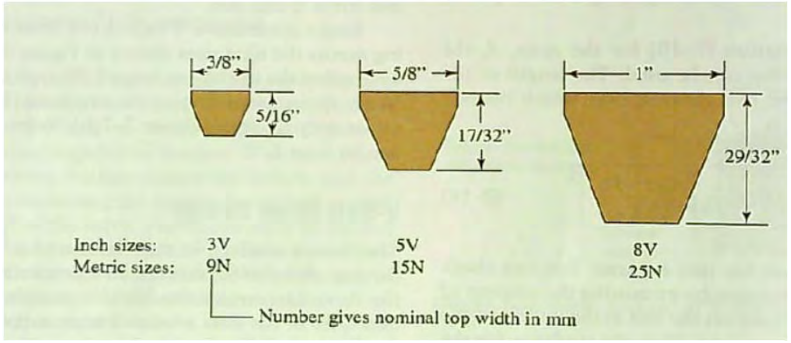
**Identify:**

$F_2$ Slack-side tension
$F_1$ Taut-side tension
$F_c$ Centrifugal tension
$(F_b)_1$ Bending tension
$(F_b)_2$ Bending tension
$F_i$ Pretension
$T_1$ Max driver tension
$T_2$ Max driven tension

# V-Belt Design (Mott)




***Heavy Duty Belts (Shigley)***




***Narrow Section V-belts (Mott)***

## Design Considerations:

- Cogged belts are better at delivering higher power at higher speeds and higher efficiency
- Plain V-Belt Max Speed Ratio 6:1
- Cogged Belts Max. Speed Ratio 7:1
- Ratio of tension between taut tension and slack tension $F_1 / F_2 \sim 5$
- Wrap angle on small sheave greater than 120º
- Use idler pulley on the slack side if center distances cannot be adjusted.
- Keep belt speeds between 1000 and 5000 fpm. 4000 fpm is a good target.

# V-Belt Design Data

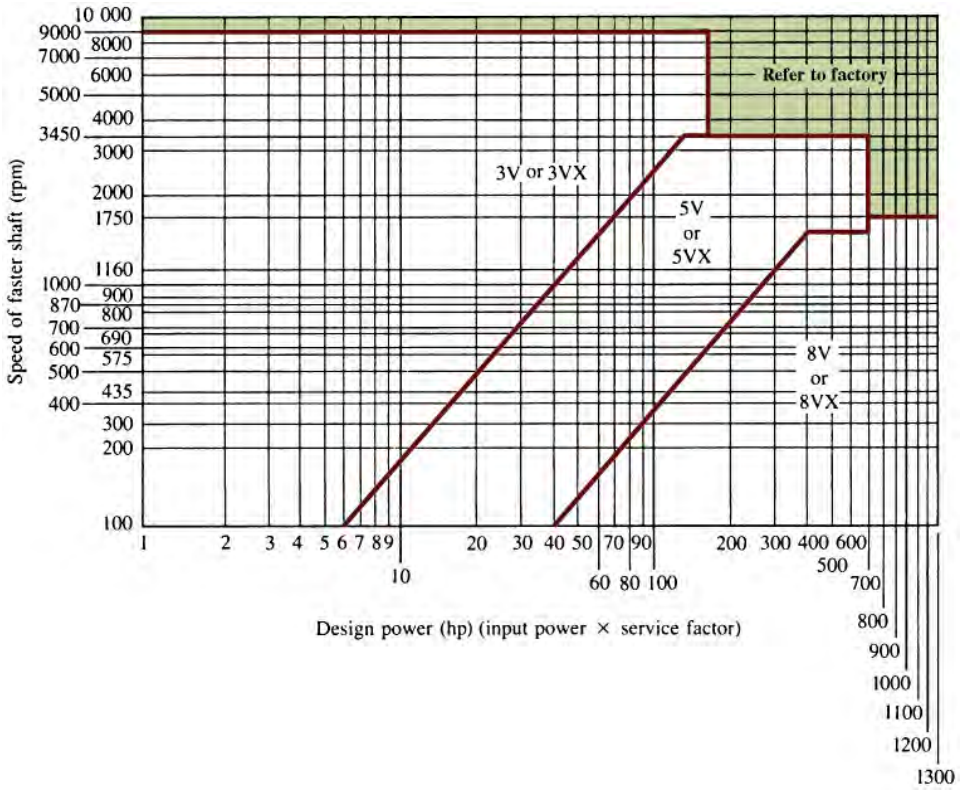
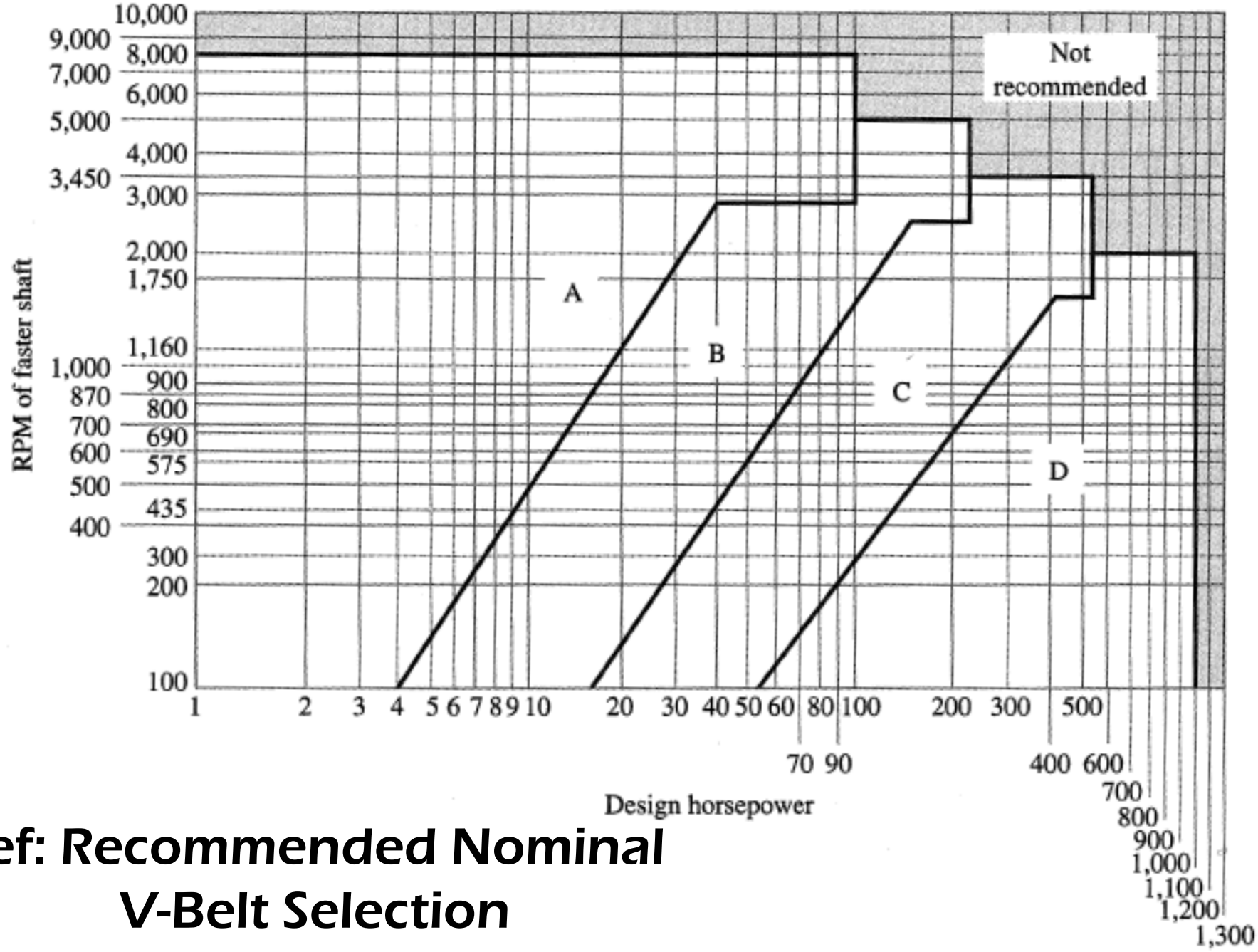



**TABLE 7–2 Standard Belt Lengths for 3V, 5V, and 8V Belts (in)**

| 3V only | 3V and 5V | 3V, 5V, and 8V | 5V and 8V | 8V only |
|---|---|---|---|---|
| 25 | 50 | 100 | 150 | 375 |
| 26.5 | 53 | 106 | 160 | 400 |
| 28 | 56 | 112 | 170 | 425 |
| 30 | 60 | 118 | 180 | 450 |
| 31.5 | 63 | 125 | 190 | 475 |
| 33.5 | 67 | 132 | 200 | 500 |
| 35.5 | 71 | 140 | 212 | |
| 37.5 | 75 | | 224 | |
| 40 | 80 | | 236 | |
| 42.5 | 85 | | 250 | |
| 45 | 90 | | 265 | |
| 47.5 | 95 | | 280 | |
| | | | 300 | |
| 165 | | | 315 | |
| | | | 335 | |
| | | | 355 | |




# Ref: Recommended Nominal V-Belt Selection

# V-Belt Design Data

- Given: Power transmitted, Input speed, Desired output speed, Application

- Compute total design power.
  - Use Table 7–1 for service factors $K_s$ .

    $P_d = P_{nom} K_s$
- Compute desired speed reduction ratio. Don't exceed 6:1 (7:1 Cogged Belts)
- Select size of V-belt using Selection Chart

**TABLE 7–1 V-Belt Service Factors[1]**

| | Driver type | | | | | |
|---|---|---|---|---|---|---|
| | AC motors: Normal torque[2] DC motors: Shunt-wound Engines: Multiple-cylinder | | | AC motors: High torque[3] DC motors: Series-wound, or compound-wound Engines: 4-cylinder or less | | |
| **Driven machine type** | **<6 h per day** | **6–15 h per day** | **>15 h per day** | **<6 h per day** | **6–15 h per day** | **>15 h per day** |
| ***Smooth loading*** | 1.0 | 1.1 | 1.2 | 1.1 | 1.2 | 1.3 |
| Agitators, light conveyors, centrifugal pumps fans and blowers under 10 hp (7.5 kW) | | | | | | |
| ***Light shock loading*** | 1.1 | 1.2 | 1.3 | 1.2 | 1.3 | 1.4 |
| Generators, machine tools mixers, fans and blowers over 10 hp (7.5 kW) gravel conveyors | | | | | | |
| ***Moderate shock loading*** | 1.2 | 1.3 | 1.4 | 1.4 | 1.5 | 1.6 |
| Bucket elevators, piston pumps textile machinery, hammer mills heavy conveyors, pulverizers | | | | | | |
| ***Heavy shock loading*** | 1.3 | 1.4 | 1.5 | 1.5 | 1.6 | 1.8 |
| Crushers, ball mills, hoists rubber mills, and extruders | | | | | | |
| ***Machinery that can choke*** | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 |

[1]Factors given are for speed reducers. For speed increases, multiply listed factors by 1.2.
[2]Synchronous, split-phase, three-phase with starting torque or breakdown torque less than 175% of full-load torque.
[3]Single-phase, three-phase with starting torque or breakdown torque greater than 175% of full-load torque.

# V-Belt Design Data

- Select size of driver sheave for required rpm and belt speed
- Estimate driven sheave diameter. Round up!
- Specify nominal center distance.

  $D_2 < CD < 3(D_2 + D_1)$
- Compute approximate belt length.

$$L_p = 2CD + 1.57(D_2 + D_1) + \frac{(D_2 - D_1)^2}{4CD} \quad (7\text{–}12)$$

- Select standard belt length from Table 7–2.
- Compute final center distance.

$$CD = \frac{B + \sqrt{B^2 - 32(D_2 - D_1)^2}}{16} \quad (7\text{–}13)$$

where $B = 4L_p - 6.28(D_2 + D_1)$

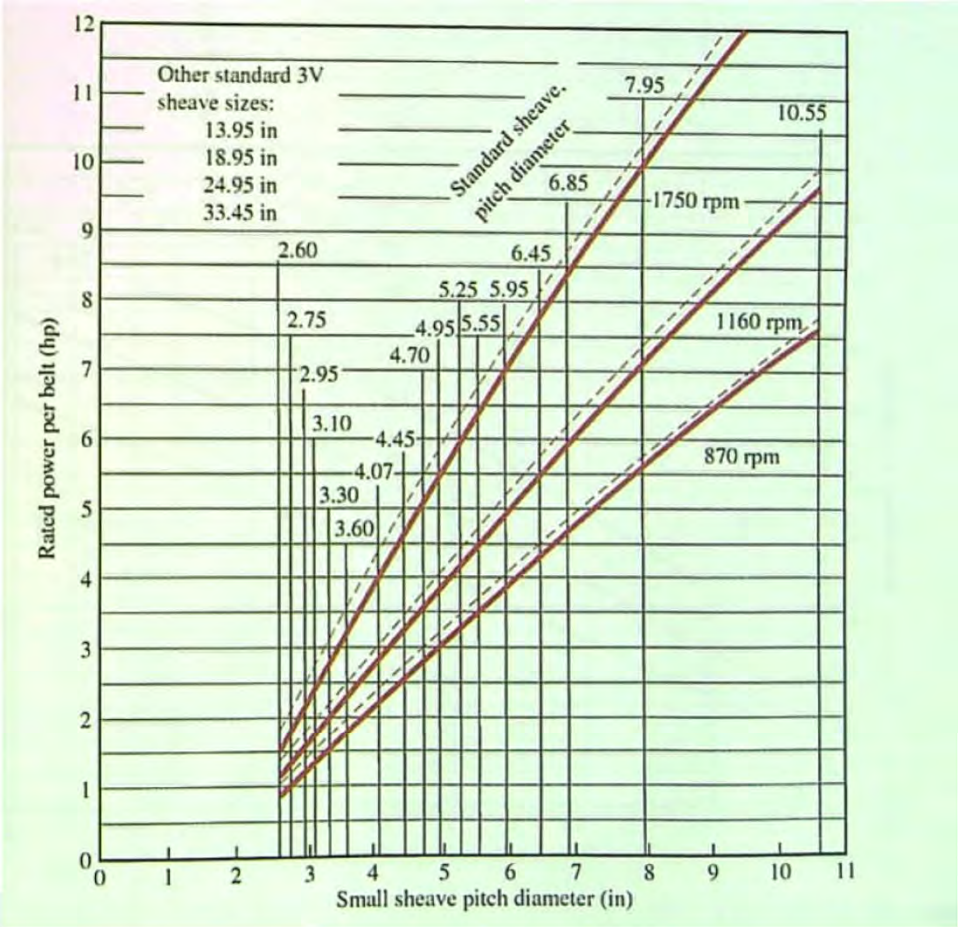



***Figure 7-14 Power Rating 3V Belts***

# V-Belt Design Data

- Compute angle of wrap on driver and driven sheaves. Compute corrected power rating for specified belt.

$$\theta_1 = 180° - 2\sin^{-1}\left[\frac{D_2 - D_1}{2CD}\right] \qquad (7\text{–}14)$$

$$\theta_2 = 180° + 2\sin^{-1}\left[\frac{D_2 - D_1}{2CD}\right] \qquad (7\text{–}15)$$

- Compute Corrected Allowable Power

$P_A = C_\theta C_L (P_{Tab} + \Delta P_R)$

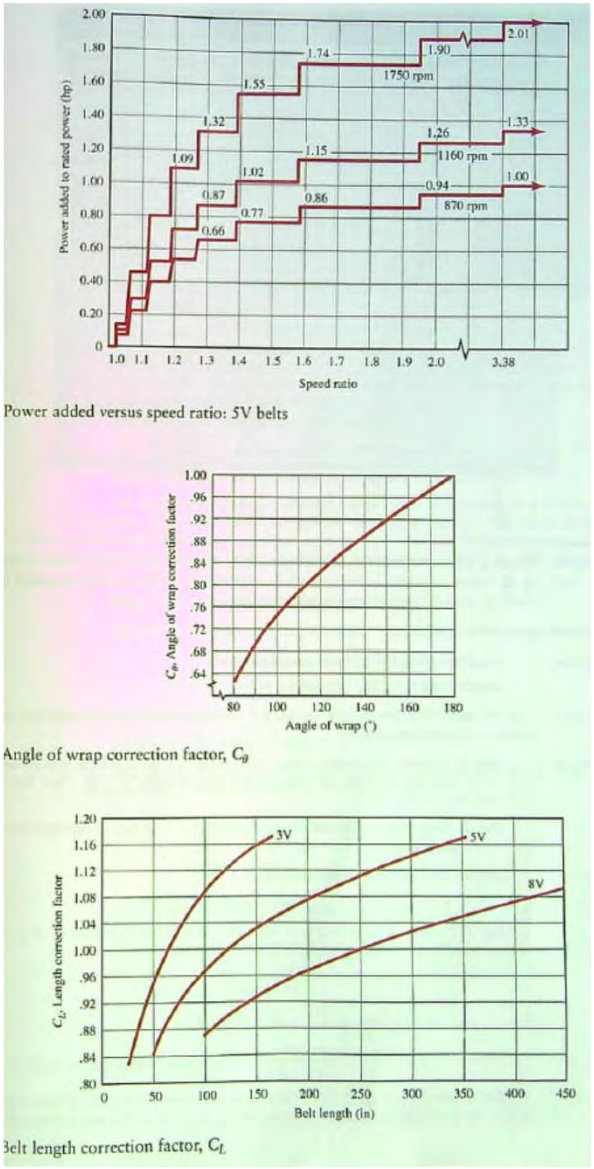
$P_{TAB}$ Design Power

$C_\theta$ Wrap Angle Correction

$C_L$ Belt Length Correction

$\Delta P_R$ Speed Ratio Power Correction

- Compute the required number of belts for total design power.



Power added versus speed ratio: 5V belts

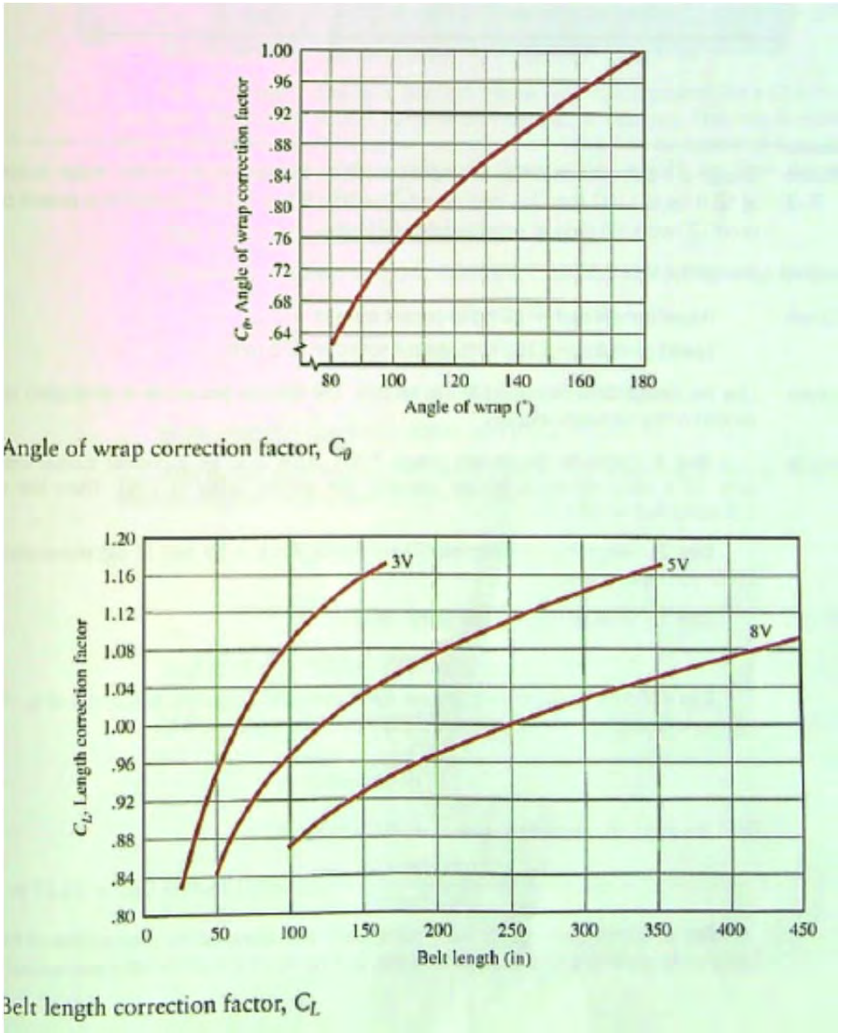
Angle of wrap correction factor, $C_\theta$

Belt length correction factor, $C_L$

## Angle of Wrap Correction Factor

| $\frac{D-d}{C}$ | $\phi$, deg | $K_1$ VV* | $K_1$ V-Flat |
|---|---|---|---|
| 0.00 | 180 | 1.00 | 0.75 |
| 0.10 | 174.3 | 0.99 | 0.76 |
| 0.20 | 166.5 | 0.97 | 0.78 |
| 0.30 | 162.7 | 0.96 | 0.79 |
| 0.40 | 156.9 | 0.94 | 0.80 |
| 0.50 | 151.0 | 0.93 | 0.81 |
| 0.60 | 145.1 | 0.91 | 0.83 |
| 0.70 | 139.0 | 0.89 | 0.84 |
| 0.80 | 132.8 | 0.87 | 0.85 |
| 0.90 | 126.5 | 0.85 | 0.85 |
| 1.00 | 120.0 | 0.82 | 0.82 |
| 1.10 | 113.3 | 0.80 | 0.80 |
| 1.20 | 106.3 | 0.77 | 0.77 |
| 1.30 | 98.9 | 0.73 | 0.73 |
| 1.40 | 91.1 | 0.70 | 0.70 |
| 1.50 | 82.8 | 0.65 | 0.65 |

Table 17–13




## Belt-Length Correction Factor

| | Nominal Belt Length, in | | | | |
|---|---|---|---|---|---|
| Length Factor | A Belts | B Belts | C Belts | D Belts | E Belts |
| 0.85 | Up to 35 | Up to 46 | Up to 75 | Up to 128 | |
| 0.90 | 38 to 46 | 48 to 60 | 81 to 96 | 144 to 162 | Up to 195 |
| 0.95 | 48 to 55 | 62 to 75 | 105 to 120 | 173 to 210 | 210 to 240 |
| 1.00 | 60 to 75 | 78 to 97 | 128 to 158 | 240 | 270 to 300 |
| 1.05 | 78 to 90 | 105 to 120 | 162 to 195 | 270 to 330 | 330 to 390 |
| 1.10 | 96 to 112 | 128 to 144 | 210 to 240 | 360 to 420 | 420 to 480 |
| 1.15 | 120 and up | 158 to 180 | 270 to 300 | 480 | 540 to 600 |
| 1.20 | | 195 and up | 330 and up | 540 and up | 660 |

*Multiply the rated horsepower per belt by this factor to obtain the corrected horsepower.

Table 17–14

**Note: Shigley does not include Speed Ratio Power Correction, $\Delta P_R$**

# Manufacturers Catalogue (Gates)

**Heavy Duty V-Belt Drive Design Manual**

## Product Features

### Super HC® V-Belts

Pioneered by Gates, these "narrow" cross-sections can transmit up to three times the horsepower of the classical cross-sections (A, B, C, and D) in the same amount of drive space.

**Markets/Applications**

Suitable for all industrial applications, particularly where space, weight and horsepower capacity are critical.

**Features/Advantages**

- **Gates Curves** provide proper cord support and full contact with the sheave-groove for uniform loading, uniform wear, and increased belt life.
- **Flex-Bonded Cords** are strongly bonded to the body of the belt resulting in equal load distribution and the absorption of bending stress without cord deterioration.
- The **Flex-Weave Cover** is a patented construction for longer cover life, providing extended protection to the core of the belt from oil, dirt, and heat.
- Meets RMA **oil and heat resistant** standards.
- Meets RMA **static conductivity** requirements.




### Super HC® Molded Notch Belts

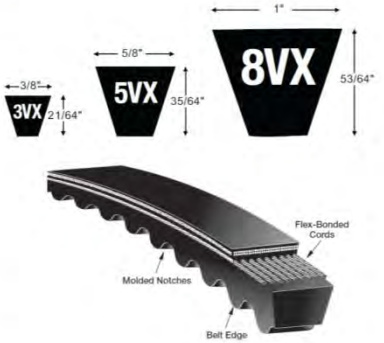
Constructed with Gates proprietary construction, this belt has a superior combination of flex and load carrying capacity, as well as transmitting more horsepower than the classical cross sections in the same amount of drive space.

**Markets/Applications**

Suitable for all industrial applications, particularly where space, weight and horsepower capacity are critical.

**Features/Advantages**

- Gates patented EPDM rubber compound technology.
- **Notches molded** into the belt during manufacturing make this belt well suited for drives with smaller diameter sheaves.
- **Belt Edge** is machined for even sheave groove contact result-ing in less slip and wear.
- **Flex-Bonded Cords** are strongly bonded to the body of the belt resulting in equal load distribution and the absorption of bending stress without cord deterioration.
- Meets RMA **oil and heat resistant** standards.
- Meets RMA **static conductivity** requirements.



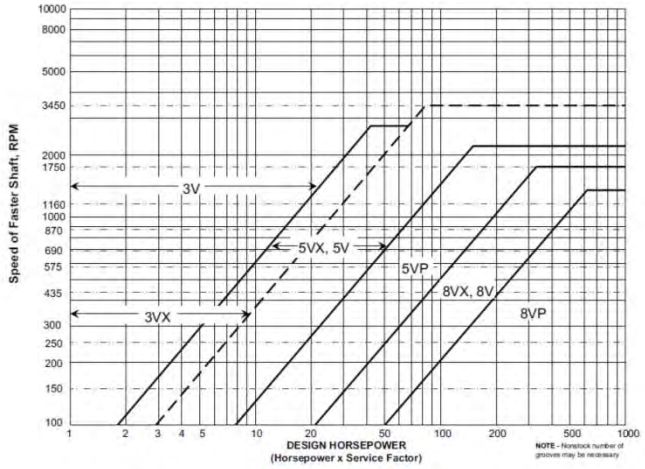




Figure No. B1

### Super HC® Molded Notch PowerBand® Belts

The PowerBand construction allows multiple belts to function as a single unit, with even load distribution and each strand fitting securely in the sheave groove.

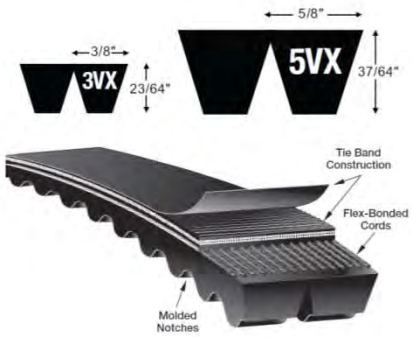
**Markets/Applications**

Recommended for multiple V-belt drives exposed to pulsating or heavy shock loads which can make belts whip, turn over or jump off the drive.

**Features/Advantages**

- Gates patented EPDM rubber compound technology.
- The **Tie Band** assures high lateral rigidity, guiding the belt in a straight line and preventing it from coming off the drive.
- **Notches molded** into the belt during manufacturing make this belt well suited for drives with smaller diameter sheaves.
- **Flex-Bonded Cords** are strongly bonded to the body of the belt resulting in equal load distribution and absorption of bending stress without cord deterioration.
- Meets RMA **oil and heat resistant** standards.
- Meets RMA **static conductivity** requirements.




## Cross Section Selection Chart

**(For Super HC® V-Belts, Super HC® Molded Notch V-Belts, Super HC PowerBand Belts, and Predator Belts)**

# Manufacturers Catalogue (Gates)

## Heavy Duty V-Belt Drive Design Manual

## Stock Drive Selection

NEMA: National Electrical Manu. Association

# NEMA Minimum Sheave Diameters

### Table No. B2

**Minimum Recommended Sheave Outside Diameters for General Purpose Electric Motors**
***Super HC® V-Belts, Super HC Molded Notch, Super HC PowerBand® Belts, Super HC Molded Notch PowerBand Belts***

(Still applies to Canada)

** For U.S. Only

| Motor Horse-power | Motor RPM (60 cycle and 50 cycle Electric Motors) 575 485* | 690 575* | 870 725* | 1160 950* | 1750 1425* | 3450 2850* | Motor Horse-power |
|---|---|---|---|---|---|---|---|
| ½ | — | — | 2.2 | — | — | — | ½ |
| ¾ | — | — | 2.4 | 2.2 | — | — | ¾ |
| 1 | 3.0 | 2.5 | 2.4 | 2.4 | 2.2 | — | 1 |
| 1½ | 3.0 | 3.0 | 2.4 | 2.4 | 2.4 | 2.2 | 1½ |
| 2 | 3.8 | 3.0 | 3.0 | 2.4 | 2.4 | 2.4 | 2 |
| 3 | 4.5 | 3.8 | 3.0 | 3.0 | 2.4 | 2.4 | 3 |
| 5 | 4.5 | 4.5 | 3.8 | 3.0 | 3.0 | 2.4 | 5 |
| 7½ | 5.2 | 4.5 | 4.4 | 3.8 | 3.0 | 3.0 | 7½ |
| 10 | 6.0 | 5.2 | 4.4 | 4.4 | 3.8 | 3.0 | 10 |
| 15 | 6.8 | 6.0 | 5.2 | 4.4 | 4.4 | 3.8 | 15 |
| 20 | 8.2 | 6.8 | 6.0 | 5.2 | 4.4 | 4.4 | 20 |
| 25 | 9.0 | 8.2 | 6.8 | 6.0 | 4.4 | 4.4 | 25 |
| 30 | 10.0 | 9.0 | 6.8 | 6.8 | 5.2 | — | 30 |
| 40 | 10.0 | 10.0 | 8.2 | 6.8 | 6.0 | — | 40 |
| 50 | 11.0 | 10.0 | 8.4 | 8.2 | 6.8 | — | 50 |
| 60 | 12.0 | 11.0 | 10.0 | 8.0 | 7.4 | — | 60 |
| 75 | 14.0 | 13.0 | 9.5 | 10.0 | 8.6 | — | 75 |
| 100 | 18.0 | 15.0 | 12.0 | 10.0 | 8.6 | — | 100 |
| 125 | 20.0 | 18.0 | 15.0 | 12.0 | 10.5# | — | 125 |
| 150 | 22.0 | 20.0 | 18.0 | 13.0 | 10.5 | — | 150 |
| 200 | 22.0 | 22.0 | 22.0 | — | 13.2 | — | 200 |
| 250 | 22.0 | 22.0 | — | — | — | — | 250 |
| 300 | 27.0 | 27.0 | — | — | — | — | 300 |

*These RPM are for 50 cycle electric motors.
#9.5 for Frame Number 444T.
Data in the white area of Table No. B2 are from NEMA Standard MG-1-14.42, November, 1978. Data in the gray area are from MG-1-14.43, January, 1968. Data in the blue area are a composite of electric motor manufacturers data. They are generally conservative, and specific motors and bearings may permit the use of a smaller motor sheave. Consult the motor manufacturer.

### Table No. B3

**Minimum Recommended Sheave Datum Diameters for General Purpose Electric Motors**
***Hi-Power® II V-Belts, Hi-Power II PowerBand Belts or Tri-Power® Molded Notch V-Belts***

** For U.S. Only

| Motor Horse-power | Motor RPM (60 cycle and 50 cycle Electric Motors) 575 485* | 690 575* | 870 725* | 1160 950* | 1750 1425* | 3450 2850* | Motor Horse-power |
|---|---|---|---|---|---|---|---|
| ½ | 2.5 | 2.5 | 2.2 | — | — | — | ½ |
| ¾ | 3.0 | 2.5 | 2.4 | 2.2 | — | — | ¾ |
| 1 | 3.0 | 3.0 | 2.4 | 2.4 | 2.2 | — | 1 |
| 1½ | 3.0 | 3.0 | 2.4 | 2.4 | 2.4 | 2.2 | 1½ |
| 2 | 3.8 | 3.0 | 3.0 | 2.4 | 2.4 | 2.4 | 2 |
| 3 | 4.5 | 3.8 | 3.0 | 3.0 | 2.4 | 2.4 | 3 |
| 5 | 4.5 | 4.5 | 3.8 | 3.0 | 3.0 | 2.6 | 5 |
| 7½ | 5.2 | 4.5 | 4.4 | 3.8 | 3.0 | 3.0 | 7½ |
| 10 | 6.0 | 5.2 | 4.6 | 4.4 | 3.8 | 3.0 | 10 |
| 15 | 6.8 | 6.0 | 5.4 | 4.6 | 4.4 | 3.8 | 15 |
| 20 | 8.2 | 6.8 | 6.0 | 5.4 | 4.6 | 4.4 | 20 |
| 25 | 9.0 | 8.2 | 6.8 | 6.0 | 5.0 | 4.4 | 25 |
| 30 | 10.0 | 9.0 | 6.8 | 6.8 | 5.4 | — | 30 |
| 40 | 10.0 | 10.0 | 8.2 | 6.8 | 6.0 | — | 40 |
| 50 | 11.0 | 10.0 | 9.0 | 8.2 | 6.8 | — | 50 |
| 60 | 12.0 | 11.0 | 10.0 | 9.0 | 7.4 | — | 60 |
| 75 | 14.0 | 13.0 | 10.5 | 10.0 | 9.0 | — | 75 |
| 100 | 18.0 | 15.0 | 12.5 | 11.0 | 10.0 | — | 100 |
| 125 | 20.0 | 18.0 | 15.0 | 12.5 | 11.5† | — | 125 |
| 150 | 22.0 | 20.0 | 18.0 | 13.0 | — | — | 150 |
| 200 | 22.0 | 22.0 | 22.0 | — | — | — | 200 |
| 250 | 22.0 | 22.0 | — | — | — | — | 250 |
| 300 | 27.0 | 27.0 | — | — | — | — | 300 |

*These RPM are for 50 cycle electric motors.
†11.0 for Frame Number 444T.
Data in the white area of Table No. B3 are from NEMA Standard MG-1-14.42, November, 1978. Data in the gray area are from MG-1-14.45, September, 1965. Data in the blue area are a composite of electric motor manufacturers data. They are generally conservative, and specific motors and bearings may permit the use of a smaller motor sheave. Consult the motor manufacturer.

**NOTE:** For a given motor horsepower and speed, the total belt pull is related to the motor sheave size. As this size decreases, the total belt pull increases. Therefore, to limit the resultant load on motor shaft and bearings, NEMA lists minimum sheave sizes for the various motors. The sheave on the motor (DriveR Sheave) should be at least this large.

# Manufacturers Catalogue (Gates)

## Table No. B10

3VX

**Example 3VX800 Belt to power an exhaust fan**
**Driver 6" sheave @1750rpm VR ~ 3.0**
**Rated Motor Power 2HP Light Shock 7hr/day**

### Rated Horsepower per Belt for 3VX Super HC® Molded Notch V-Belts and Super HC Molded Notch PowerBand® Belts

$H_{Tab}$

| RPM of Faster Shaft | Basic Horsepower per Belt for Small Sheave Outside Diameter | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2.20 | 2.35 | 2.50 | 2.65 | 2.80 | 3.00 | 3.15 | 3.35 | 3.65 | 4.12 | 4.50 | 4.75 | 5.00 | 5.30 | 5.60 | 6.00 | 6.50 | 6.90 | 8.00 | 10.6 |
| 575 | 0.55 | 0.64 | 0.73 | 0.83 | 0.92 | 1.04 | 1.13 | 1.25 | 1.43 | 1.72 | 1.94 | 2.09 | 2.24 | 2.41 | 2.59 | 2.82 | 3.11 | 3.34 | 3.97 | 5.42 |
| 690 | 0.64 | 0.75 | 0.86 | 0.97 | 1.08 | 1.22 | 1.33 | 1.48 | 1.69 | 2.02 | 2.29 | 2.47 | 2.64 | 2.85 | 3.06 | 3.33 | 3.67 | 3.95 | 4.69 | 6.39 |
| 725 | 0.66 | 0.78 | 0.90 | 1.01 | 1.13 | 1.28 | 1.39 | 1.54 | 1.77 | 2.12 | 2.40 | 2.58 | 2.76 | 2.98 | 3.20 | 3.49 | 3.84 | 4.13 | 4.90 | 6.69 |
| 870 | 0.77 | 0.91 | 1.05 | 1.18 | 1.32 | 1.50 | 1.63 | 1.81 | 2.08 | 2.49 | 2.82 | 3.04 | 3.26 | 3.51 | 3.77 | 4.11 | 4.53 | 4.87 | 5.78 | 7.87 |
| 950 | 0.83 | 0.98 | 1.13 | 1.28 | 1.42 | 1.62 | 1.77 | 1.96 | 2.25 | 2.70 | 3.06 | 3.29 | 3.52 | 3.80 | 4.08 | 4.45 | 4.91 | 5.27 | 6.25 | 8.51 |
| 1160 | 0.98 | 1.16 | 1.34 | 1.52 | 1.69 | 1.93 | 2.10 | 2.34 | 2.68 | 3.22 | 3.65 | 3.93 | 4.21 | 4.55 | 4.88 | 5.32 | 5.87 | 6.30 | 7.47 | 10.1 |
| 1425 | 1.16 | 1.38 | 1.59 | 1.81 | 2.02 | 2.31 | 2.52 | 2.80 | 3.22 | 3.86 | 4.38 | 4.72 | 5.06 | 5.46 | 5.86 | 6.38 | 7.03 | 7.55 | 8.94 | 12.1 |
| 1750 | 1.37 | 1.63 | 1.89 | 2.15 | 2.41 | 2.75 | 3.01 | 3.34 | 3.85 | 4.63 | 5.25 | 5.65 | 6.06 | 6.53 | 7.01 | 7.63 | 8.40 | 9.01 | 10.6 | 14.2 |
| 2850 | 2.00 | 2.41 | 2.81 | 3.21 | 3.61 | 4.14 | 4.53 | 5.05 | 5.82 | 6.99 | 7.92 | 8.53 | 9.12 | 9.82 | 10.5 | 11.4 | 12.5 | 13.3 | 15.5 | |
| 3450 | 2.30 | 2.78 | 3.26 | 3.74 | 4.21 | 4.82 | 5.28 | 5.89 | 6.78 | 8.15 | 9.21 | 9.90 | 10.6 | 11.4 | 12.1 | 13.1 | 14.3 | 15.2 | | |
| 200 | 0.22 | 0.26 | 0.29 | 0.33 | 0.36 | 0.41 | 0.44 | 0.48 | 0.55 | 0.66 | 0.74 | 0.80 | 0.85 | 0.92 | 0.98 | 1.07 | 1.18 | 1.27 | 1.50 | 2.05 |
| 400 | 0.40 | 0.47 | 0.54 | 0.60 | 0.67 | 0.75 | 0.82 | 0.91 | 1.03 | 1.24 | 1.40 | 1.50 | 1.61 | 1.73 | 1.86 | 2.03 | 2.23 | 2.40 | 2.85 | 3.89 |
| 600 | 0.57 | 0.66 | 0.76 | 0.86 | 0.95 | 1.08 | 1.18 | 1.30 | 1.49 | 1.78 | 2.02 | 2.17 | 2.33 | 2.51 | 2.69 | 2.93 | 3.23 | 3.47 | 4.13 | 5.63 |
| 800 | 0.72 | 0.85 | 0.98 | 1.10 | 1.23 | 1.39 | 1.52 | 1.68 | 1.93 | 2.31 | 2.62 | 2.82 | 3.02 | 3.26 | 3.50 | 3.81 | 4.20 | 4.51 | 5.36 | 7.30 |
| 1000 | 0.87 | 1.02 | 1.18 | 1.34 | 1.49 | 1.69 | 1.85 | 2.05 | 2.35 | 2.82 | 3.20 | 3.45 | 3.69 | 3.98 | 4.27 | 4.66 | 5.14 | 5.52 | 6.55 | 8.90 |
| 1200 | 1.01 | 1.19 | 1.38 | 1.56 | 1.74 | 1.99 | 2.17 | 2.41 | 2.76 | 3.32 | 3.76 | 4.05 | 4.34 | 4.69 | 5.03 | 5.48 | 6.04 | 6.49 | 7.69 | 10.4 |
| 1400 | 1.14 | 1.36 | 1.57 | 1.78 | 1.99 | 2.27 | 2.48 | 2.75 | 3.17 | 3.80 | 4.32 | 4.65 | 4.98 | 5.37 | 5.77 | 6.29 | 6.93 | 7.43 | 8.80 | 11.9 |
| 1600 | 1.27 | 1.52 | 1.76 | 1.99 | 2.23 | 2.55 | 2.78 | 3.09 | 3.56 | 4.28 | 4.85 | 5.23 | 5.60 | 6.04 | 6.48 | 7.06 | 7.78 | 8.35 | 9.87 | 13.2 |
| 1800 | 1.40 | 1.67 | 1.94 | 2.20 | 2.47 | 2.82 | 3.08 | 3.43 | 3.94 | 4.74 | 5.38 | 5.79 | 6.21 | 6.70 | 7.18 | 7.82 | 8.61 | 9.23 | 10.9 | 14.5 |
| 2000 | 1.52 | 1.82 | 2.11 | 2.40 | 2.70 | 3.08 | 3.37 | 3.75 | 4.32 | 5.19 | 5.89 | 6.34 | 6.79 | 7.33 | 7.86 | 8.55 | 9.41 | 10.1 | 11.9 | 15.7 |
| 2200 | 1.64 | 1.96 | 2.28 | 2.60 | 2.92 | 3.34 | 3.65 | 4.07 | 4.68 | 5.63 | 6.39 | 6.88 | 7.37 | 7.94 | 8.51 | 9.26 | 10.2 | 10.9 | 12.8 | 16.8 |
| 2400 | 1.75 | 2.10 | 2.45 | 2.80 | 3.14 | 3.59 | 3.93 | 4.38 | 5.04 | 6.06 | 6.88 | 7.40 | 7.92 | 8.54 | 9.15 | 9.95 | 10.9 | 11.7 | 13.7 | |
| 2600 | 1.87 | 2.24 | 2.61 | 2.98 | 3.35 | 3.84 | 4.20 | 4.68 | 5.39 | 6.48 | 7.35 | 7.91 | 8.47 | 9.12 | 9.77 | 10.6 | 11.6 | 12.4 | 14.5 | |
| 2800 | 1.97 | 2.37 | 2.77 | 3.17 | 3.56 | 4.08 | 4.47 | 4.98 | 5.73 | 6.89 | 7.81 | 8.40 | 8.99 | 9.68 | 10.4 | 11.2 | 12.3 | 13.1 | 15.3 | |
| 3000 | 2.08 | 2.50 | 2.93 | 3.35 | 3.76 | 4.31 | 4.72 | 5.26 | 6.06 | 7.29 | 8.26 | 8.88 | 9.50 | 10.2 | 10.9 | 11.8 | 13.0 | 13.8 | 16.0 | |
| 3200 | 2.18 | 2.63 | 3.08 | 3.52 | 3.96 | 4.54 | 4.98 | 5.55 | 6.39 | 7.68 | 8.69 | 9.34 | 9.99 | 10.7 | 11.5 | 12.4 | 13.6 | 14.4 | | |
| 3400 | 2.28 | 2.75 | 3.23 | 3.69 | 4.16 | 4.77 | 5.22 | 5.82 | 6.71 | 8.05 | 9.11 | 9.79 | 10.5 | 11.2 | 12.0 | 13.0 | 14.1 | 15.0 | | |
| 3600 | 2.37 | 2.87 | 3.37 | 3.86 | 4.35 | 4.99 | 5.46 | 6.09 | 7.01 | 8.42 | 9.52 | 10.2 | 10.9 | 11.7 | 12.5 | 13.5 | 14.7 | | | |
| 3800 | 2.47 | 2.99 | 3.51 | 4.02 | 4.53 | 5.20 | 5.70 | 6.35 | 7.31 | 8.77 | 9.91 | 10.6 | 11.3 | 12.2 | 13.0 | 14.0 | 15.2 | | | |
| 4000 | 2.56 | 3.10 | 3.65 | 4.18 | 4.71 | 5.41 | 5.92 | 6.60 | 7.60 | 9.11 | 10.3 | 11.0 | 11.8 | 12.6 | 13.4 | 14.4 | | | | |
| 4200 | 2.64 | 3.21 | 3.78 | 4.33 | 4.89 | 5.61 | 6.15 | 6.85 | 7.88 | 9.44 | 10.6 | 11.4 | 12.1 | 13.0 | 13.8 | | | | | |
| 4400 | 2.73 | 3.32 | 3.91 | 4.48 | 5.06 | 5.81 | 6.36 | 7.09 | 8.15 | 9.75 | 11.0 | 11.8 | 12.5 | 13.4 | 14.2 | | | | | |
| 4600 | 2.81 | 3.42 | 4.03 | 4.63 | 5.22 | 6.00 | 6.57 | 7.32 | 8.41 | 10.1 | 11.3 | 12.1 | 12.9 | 13.7 | | | | | | |
| 4800 | 2.89 | 3.52 | 4.15 | 4.77 | 5.38 | 6.18 | 6.77 | 7.54 | 8.66 | 10.3 | 11.6 | 12.4 | 13.2 | | | | | | | |
| 5000 | 2.96 | 3.62 | 4.27 | 4.91 | 5.53 | 6.36 | 6.96 | 7.75 | 8.90 | 10.6 | 11.9 | 12.7 | | | | | | | | |
| 5200 | 3.03 | 3.71 | 4.38 | 5.04 | 5.68 | 6.53 | 7.15 | 7.96 | 9.13 | 10.9 | 12.2 | 13.0 | | | | | | | | |
| 5400 | 3.10 | 3.80 | 4.49 | 5.16 | 5.83 | 6.69 | 7.33 | 8.15 | 9.35 | 11.1 | 12.4 | | | | | | | | | |
| 5600 | 3.17 | 3.89 | 4.59 | 5.28 | 5.96 | 6.85 | 7.50 | 8.34 | 9.56 | 11.3 | | | | | | | | | | |
| 5800 | 3.23 | 3.97 | 4.69 | 5.40 | 6.10 | 7.00 | 7.66 | 8.52 | 9.76 | 11.6 | | | | | | | | | | |
| 6000 | 3.29 | 4.05 | 4.79 | 5.51 | 6.22 | 7.15 | 7.82 | 8.69 | 9.94 | 11.7 | | | | | | | | | | |
| 6200 | 3.35 | 4.12 | 4.88 | 5.62 | 6.34 | 7.28 | 7.97 | 8.85 | 10.1 | | | | | | | | | | | |
| 6400 | 3.40 | 4.19 | 4.96 | 5.72 | 6.46 | 7.41 | 8.11 | 9.00 | 10.3 | | | | | | | | | | | |

Drives for rpm-diameter combinations where no horsepower is shown may be practical if all conditions are known. See your local Gates representative

| RPM of Faster Shaft | Additional Horsepower per Belt for Speed Ratio | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 1.00 to 1.01 | 1.02 to 1.03 | 1.04 to 1.06 | 1.07 to 1.09 | 1.10 to 1.13 | 1.14 to 1.18 | 1.19 to 1.25 | 1.26 to 1.35 | 1.36 to 1.57 | 1.58 and over |
| 575 | 0.00 | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 | 0.05 | 0.06 | 0.07 | 0.08 |
| 690 | 0.00 | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 | 0.07 | 0.08 | 0.09 | 0.10 |
| 725 | 0.00 | 0.01 | 0.02 | 0.03 | 0.05 | 0.06 | 0.07 | 0.08 | 0.09 | 0.10 |
| 870 | 0.00 | 0.01 | 0.03 | 0.04 | 0.05 | 0.07 | 0.08 | 0.10 | 0.11 | 0.12 |
| 950 | 0.00 | 0.01 | 0.03 | 0.04 | 0.06 | 0.07 | 0.09 | 0.10 | 0.12 | 0.13 |
| 1160 | 0.00 | 0.02 | 0.04 | 0.05 | 0.07 | 0.09 | 0.11 | 0.13 | 0.15 | 0.16 |
| 1425 | 0.00 | 0.02 | 0.04 | 0.07 | 0.09 | 0.11 | 0.13 | 0.16 | 0.18 | 0.20 |
| 1750 | 0.00 | 0.03 | 0.06 | 0.08 | 0.11 | 0.14 | 0.17 | 0.19 | 0.22 | 0.25 |
| 2850 | 0.00 | 0.04 | 0.09 | 0.13 | 0.18 | 0.22 | 0.27 | 0.31 | 0.36 | 0.40 |
| 3450 | 0.00 | 0.05 | 0.11 | 0.16 | 0.22 | 0.27 | 0.33 | 0.38 | 0.43 | 0.49 |
| 200 | 0.00 | 0.00 | 0.01 | 0.01 | 0.01 | 0.02 | 0.02 | 0.02 | 0.03 | 0.03 |
| 400 | 0.00 | 0.01 | 0.01 | 0.02 | 0.03 | 0.03 | 0.04 | 0.04 | 0.05 | 0.06 |
| 600 | 0.00 | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 | 0.06 | 0.07 | 0.08 | 0.08 |
| 800 | 0.00 | 0.01 | 0.03 | 0.04 | 0.05 | 0.06 | 0.08 | 0.09 | 0.10 | 0.11 |
| 1000 | 0.00 | 0.02 | 0.03 | 0.05 | 0.06 | 0.08 | 0.09 | 0.11 | 0.13 | 0.14 |
| 1200 | 0.00 | 0.02 | 0.04 | 0.06 | 0.08 | 0.09 | 0.11 | 0.13 | 0.15 | 0.17 |
| 1400 | 0.00 | 0.02 | 0.04 | 0.07 | 0.09 | 0.11 | 0.13 | 0.15 | 0.18 | 0.20 |
| 1600 | 0.00 | 0.03 | 0.05 | 0.08 | 0.10 | 0.13 | 0.15 | 0.18 | 0.20 | 0.23 |
| 1800 | 0.00 | 0.03 | 0.06 | 0.08 | 0.11 | 0.14 | 0.17 | 0.20 | 0.23 | 0.25 |
| 2000 | 0.00 | 0.03 | 0.06 | 0.09 | 0.13 | 0.16 | 0.19 | 0.22 | 0.25 | 0.28 |
| 2200 | 0.00 | 0.03 | 0.07 | 0.10 | 0.14 | 0.17 | 0.21 | 0.24 | 0.28 | 0.31 |
| 2400 | 0.00 | 0.04 | 0.08 | 0.11 | 0.15 | 0.19 | 0.23 | 0.26 | 0.30 | 0.34 |
| 2600 | 0.00 | 0.04 | 0.08 | 0.12 | 0.16 | 0.20 | 0.25 | 0.29 | 0.33 | 0.37 |
| 2800 | 0.00 | 0.04 | 0.09 | 0.13 | 0.18 | 0.22 | 0.26 | 0.31 | 0.35 | 0.40 |
| 3000 | 0.00 | 0.05 | 0.09 | 0.14 | 0.19 | 0.24 | 0.28 | 0.33 | 0.38 | 0.42 |
| 3200 | 0.00 | 0.05 | 0.10 | 0.15 | 0.20 | 0.25 | 0.30 | 0.35 | 0.40 | 0.45 |
| 3400 | 0.00 | 0.05 | 0.11 | 0.16 | 0.21 | 0.27 | 0.32 | 0.37 | 0.43 | 0.48 |
| 3600 | 0.00 | 0.06 | 0.11 | 0.17 | 0.23 | 0.28 | 0.34 | 0.40 | 0.45 | 0.51 |
| 3800 | 0.00 | 0.06 | 0.12 | 0.18 | 0.24 | 0.30 | 0.36 | 0.42 | 0.48 | 0.54 |
| 4000 | 0.00 | 0.06 | 0.13 | 0.19 | 0.25 | 0.31 | 0.38 | 0.44 | 0.50 | 0.57 |
| 4200 | 0.00 | 0.07 | 0.13 | 0.20 | 0.26 | 0.33 | 0.40 | 0.46 | 0.53 | 0.59 |
| 4400 | 0.00 | 0.07 | 0.14 | 0.21 | 0.28 | 0.35 | 0.42 | 0.48 | 0.55 | 0.62 |
| 4600 | 0.00 | 0.07 | 0.14 | 0.22 | 0.29 | 0.36 | 0.43 | 0.51 | 0.58 | 0.65 |
| 4800 | 0.00 | 0.08 | 0.15 | 0.23 | 0.30 | 0.38 | 0.45 | 0.53 | 0.60 | 0.68 |
| 5000 | 0.00 | 0.08 | 0.16 | 0.24 | 0.31 | 0.39 | 0.47 | 0.55 | 0.63 | 0.71 |
| 5200 | 0.00 | 0.08 | 0.16 | 0.25 | 0.33 | 0.41 | 0.49 | 0.57 | 0.65 | 0.74 |
| 5400 | 0.00 | 0.09 | 0.17 | 0.25 | 0.34 | 0.42 | 0.51 | 0.59 | 0.68 | 0.76 |
| 5600 | 0.00 | 0.09 | 0.18 | 0.26 | 0.35 | 0.44 | 0.53 | 0.62 | 0.71 | 0.79 |
| 5800 | 0.00 | 0.09 | 0.18 | 0.27 | 0.36 | 0.46 | 0.55 | 0.64 | 0.73 | 0.82 |
| 6000 | 0.00 | 0.09 | 0.19 | 0.28 | 0.38 | 0.47 | 0.57 | 0.66 | 0.76 | 0.85 |
| 6200 | 0.00 | 0.10 | 0.20 | 0.29 | 0.39 | 0.49 | 0.59 | 0.68 | 0.78 | 0.88 |
| 6400 | 0.00 | 0.10 | 0.20 | 0.30 | 0.40 | 0.50 | 0.60 | 0.70 | 0.81 | 0.91 |

$\Delta P_R$

Heavy Duty V-Belt Drive Design Manual

# Ref: Belt Service Factors ($K_s$)

| Driven Equipment | Operating Period /day | | | | | |
|---|---|---|---|---|---|---|
| | < 8 h | < 16 h | > 16 h | < 8 h | < 16 h | >16 h |
| | **Driving Machine:** Reasonable Smooth Starting torque 100-200% FL | | | **Driving Machine:** Coarse Starting torque 200-400% | | |
| | DC motor (shunt wound), AC motor (star-delta), IC engines 8+ Cyl., Turbines | | | DC motor (series / mixed excitation), AC motors, 2 stroke engines, IC engine <8 Cyl. or under 600 rpm | | |
| Smooth Machine Tools including lathes, drills, etc. Light blowers, Light-duty Conveyors, Centrifugal pumps | 1.0 | 1.1 | 1.2 | 1.1 | 1.2 | 1.3 |
| Uneven Machine Tools including milling, gearcutting, etc. Printing machines, Laundry machines, Fans, Punches/Presses, Generators, Piston pumps and compressors (4+ Cyl.), Chain conveyors, Elevators, Transmissions, High-speed grinders | 1.1 | 1.2 | 1.3 | 1.2 | 1.3 | 1.4 |
| Planing machines, piston pumps and compressors (1 or 2 Cyl.), Heavy fans and blowers, Exciters, Crushers, Pelletizing machines | 1.2 | 1.3 | 1.4 | 1.3 | 1.4 | 1.6 |
| Hoists, Excavators, Heavy Presses, Mechanical Hammers, Mills, Stone crushers, Extruders, Sawmill machines, Dredgers | 1.3 | 1.4 | 1.5 | 1.5 | 1.6 | 1.8 |

## Table No. B4

# Super HC® and Super HC Molded Notch V-Belts and PowerBand® Belts Sizes

(PowerBand Belts are available in 2, 3, 4 or 5 bands in sizes shown, or wider, on a standard non-stock basis.)




Lengths listed as molded notch are available in banded or molded notch construction unless otherwise noted.

| 3V Part No. | Outside Circum. Effective Length (in) | 3V Part No. | Outside Circum. Effective Length (in) | 5V Part No. | Outside Circum. Effective Length (in) | 5V Part No. | Outside Circum. Effective Length (in) | 5V Part No. | Outside Circum. Effective Length (in) | 8V Part No. | Outside Circum. Effective Length (in) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3VX250* | 25 | 3VX690** | 69 | 5VX350* | 35 | 5VX720* | 72 | 5VX1160* | 116 | 8V1000* | 100 |
| 3VX265* | 26.5 | 3VX710 | 71 | 5VX362* | 36.2 | 5VX730* | 73 | 5VX1162* | 116.2 | 8V1060* | 106 |
| 3VX280* | 28 | 3V730* | 73 | 5VX372* | 37.2 | 5VX740* | 74 | 5VX1180** | 118 | 8V1120* | 112 |
| 3VX290** | 29 | 3VX750 | 75 | 5VX382* | 38.2 | 5VX750 | 75 | 5V1200** | 120 | 8V1180* | 118 |
| 3VX300 | 30 | 3VX771** | 77.1 | 5VX392* | 39.2 | 5VX760* | 76 | 5V1210** | 121 | 8V1250* | 125 |
| 3VX315 | 31.5 | 3VX800 | 80 | 5VX402* | 40.2 | 5VX769* | 76.9 | 5VX1220* | 122 | 8V1320* | 132 |
| 3VX326** | 32.6 | 3V810* | 81 | 5VX412* | 41.2 | 5VX780* | 78 | 5VX1230* | 123 | 8V1400* | 140 |
| 3VX335 | 33.5 | 3VX826** | 82.6 | 5VX422* | 42.2 | 5VX790* | 79 | 5VX1250 | 125 | 8V1500* | 150 |
| 3VX350** | 35 | 3V830* | 83 | 5VX433* | 43.3 | 5VX800 | 80 | 5VX1277* | 127.7 | 8V1600* | 160 |
| 3VX355 | 35.5 | 3VX850 | 85 | 5VX450* | 45 | 5VX810* | 81 | 5VX1320 | 132 | 8V1700* | 170 |
| 3VX366** | 36.6 | 3VX900 | 90 | 5VX459* | 45.9 | 5VX830* | 83 | 5VX1374* | 137.4 | 8V1800* | 180 |
| 3VX375 | 37.5 | 3VX926** | 92.6 | 5VX470* | 47 | 5VX840* | 84 | 5VX1400 | 140 | 8V1900* | 190 |
| 3VX385** | 38.5 | 3VX950 | 95 | 5VX479* | 47.9 | 5VX850 | 85 | 5VX1469* | 146.9 | 8V2000* | 200 |
| 3VX390** | 39 | 3VX974** | 97.4 | 5VX490* | 49 | 5VX860* | 86 | 5VX1500 | 150 | 8V2120 | 212 |
| 3VX400 | 40 | 3VX1000 | 100 | 5VX500 | 50 | 5VX867* | 86.7 | 5VX1600 | 160 | 8V2240 | 224 |
| 3VX415** | 41.5 | 3VX1027** | 102.7 | 5VX510* | 51 | 5VX880* | 88 | 5VX1700 | 170 | 8V2300** | 230 |
| 3VX425 | 42.5 | 3VX1060 | 106 | 5VX519* | 51.9 | 5VX890* | 89 | 5VX1701* | 170.1 | 8V2360 | 236 |
| 3VX450 | 45 | 3VX1088** | 108.8 | 5VX530 | 53 | 5VX900 | 90 | 5VX1800 | 180 | 8V2500 | 250 |
| 3VX464** | 46.4 | 3VX1120 | 112 | 5VX540* | 54 | 5VX918* | 91.8 | 5VX1900 | 190 | 8V2650 | 265 |
| 3VX475 | 47.5 | 3VX1146** | 114.6 | 5VX550* | 55 | 5VX930* | 93 | 5VX2000 | 200 | 8V2800 | 280 |
| 3VX487** | 48.7 | 3VX1180 | 118 | 5VX560 | 56 | 5VX940* | 94 | 5V1630*** | 163 | 8V3000 | 300 |
| 3VX500 | 50 | 3VX1224** | 122.4 | 5VX570* | 57 | 5VX950 | 95 | 5V2120 | 212 | 8V3150 | 315 |
| 3VX520** | 52 | 3VX1250 | 125 | 5VX580* | 58 | 5VX960* | 96 | 5V2240 | 224 | 8V3350 | 335 |
| 3VX530 | 53 | 3VX1296** | 129.6 | 5VX590* | 59 | 5VX978* | 97.8 | 5V2360 | 236 | 8V3550 | 355 |
| 3VX540** | 54 | 3VX1320 | 132 | 5VX600 | 60 | 5VX990* | 99 | 5V2500 | 250 | 8V3750 | 375 |
| 3VX550** | 55 | 3VX1400 | 140 | 5VX610* | 61 | 5VX1000 | 100 | 5V2650 | 265 | 8V4000 | 400 |
| 3VX560 | 56 | | | 5VX619* | 61.9 | 5VX1017* | 101.7 | 5V2800 | 280 | 8V4250 | 425 |
| 3VX570** | 57 | | | 5VX630 | 63 | 5VX1030* | 103 | 5V3000 | 300 | 8V4500 | 450 |
| 3VX580** | 58 | | | 5VX650* | 65 | 5VX1050* | 105 | 5V3150 | 315 | 8V4750 | 475 |
| 3VX590** | 59 | | | 5VX660* | 66 | 5VX1060 | 106 | 5V3350 | 335 | 8V5000 | 500 |
| 3VX600 | 60 | | | 5VX670 | 67 | 5VX1080* | 108 | 5V3550 | 355 | 8V5600 | 560 |
| 3VX616** | 61.6 | | | 5VX680* | 68 | 5VX1108* | 110.8 | | | 8V6000 | 600 |
| 3VX630 | 63 | | | 5VX690* | 69 | 5VX1120 | 112 | | | | |
| 3VX650*/*** | 65 | | | 5VX700* | 70 | 5VX1139* | 113.9 | | | | |
| 3VX670 | 67 | | | 5VX710 | 71 | 5VX1150* | 115 | | | | |

* Not Available in 3V PowerBand
** Only Available in 3VX Single Belts
*** Not Available in 3VX PowerBand

* Only Available in 5VX Single Belt
** Only Available in 5V PowerBand
*** Only Available in 5V Single Belt

* Available in 8VX Single Belt
** Only Available in 8V Single Belt

## Table No. B6

# Super HC® V-Belt, Super HC Molded Notch® V-Belt, Predator® V-Belt, Super HC PowerBand® Belt, Super HC Molded Notch PowerBand Belt and Predator PowerBand Belt Drives




| Sheave Outside Diameters | | | V-Belt No. and Center Distance | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 3V 3VX | 3V 3VX | 3V 3VX | 3VX | 3V 3VX | 3V 3VX | 3VX | 3V 3VX | 3VX | 3V 3VX | 3V 3VX | 3V 3VX | 3VX | 3VX | 3V 3VX | 3VX | 3V 3VX | 3V 3VX 3VP | 3VX | 3V 3VX 3VP | 3VX | 3V 3VX 3VP | 3VX | 3V 3VX 3VP | 3VX |
| Small Sheave | Large Sheave | Speed Ratio | 250 | 265 | 280 | 290 | 300 | 315 | 326 | 335 | 350 | 355 | 366 | 375 | 385 | 390 | 400 | 415 | 425 | 450 | 464 | 475 | 487 | 500 | 520 | 530 | 540 |
| * 2.50 | 4.75 | 1.92 | 6.7 | 7.5 | 8.2 | 8.7 | 9.2 | 10.0 | 10.5 | 11.0 | 11.8 | 12.0 | 12.6 | 13.0 | 13.5 | 13.8 | 14.3 | 15.0 | 15.5 | 16.8 | 17.5 | 18.0 | 18.6 | 19.3 | 20.3 | 20.8 | 21.3 |
| 3.15 | 6.00 | 1.92 | 5.1 | 5.9 | 6.7 | 7.2 | 7.7 | 8.4 | 9.0 | 9.5 | 10.2 | 10.5 | 11.0 | 11.5 | 12.0 | 12.2 | 12.7 | 13.5 | 14.0 | 15.2 | 15.9 | 16.5 | 17.1 | 17.8 | 18.8 | 19.3 | 19.8 |
| * 2.35 | 4.50 | 1.93 | 7.0 | 7.8 | 8.6 | 9.1 | 9.6 | 10.3 | 10.9 | 11.3 | 12.1 | 12.3 | 12.9 | 13.3 | 13.8 | 14.1 | 14.6 | 15.3 | 15.8 | 17.1 | 17.8 | 18.3 | 18.9 | 19.6 | 20.6 | 21.1 | 21.6 |
| 3.35 | 6.50 | 1.95 | | | 6.1 | 6.6 | 7.1 | 7.9 | 8.4 | 8.9 | 9.5 | 9.9 | 10.4 | 10.9 | 11.4 | 11.7 | 12.2 | 12.9 | 13.4 | 14.7 | 15.4 | 15.9 | 16.5 | 17.2 | 18.2 | 18.7 | 19.2 |
| 4.12 | 8.00 | 1.95 | | | | | | | | 7.0 | 7.7 | 8.0 | 8.6 | 9.0 | 9.5 | 9.8 | 10.3 | 11.1 | 11.6 | 12.8 | 13.5 | 14.1 | 14.7 | 15.4 | 16.4 | 16.9 | 17.4 |
| 5.30 | 10.60 | 2.01 | | | | | | | | | | | | | | | | | | 9.6 | 10.4 | 10.9 | 11.6 | 12.2 | 13.2 | 13.8 | 14.3 |
| * 2.50 | 5.00 | 2.02 | 6.5 | 7.3 | 8.0 | 8.5 | 9.0 | 9.8 | 10.3 | 10.8 | 11.5 | 11.8 | 12.3 | 12.8 | 13.3 | 13.6 | 14.1 | 14.8 | 15.3 | 16.6 | 17.3 | 17.8 | 18.4 | 19.1 | 20.1 | 20.6 | 21.1 |
| 2.65 | 5.30 | 2.02 | 6.1 | 6.9 | 7.6 | 8.1 | 8.7 | 9.4 | 10.0 | 10.4 | 11.2 | 11.4 | 12.0 | 12.4 | 12.9 | 13.2 | 13.7 | 14.4 | 14.9 | 16.2 | 16.9 | 17.5 | 18.1 | 18.7 | 19.7 | 20.2 | 20.7 |
| 2.80 | 5.60 | 2.02 | 5.7 | 6.5 | 7.3 | 7.8 | 8.3 | 9.0 | 9.6 | 10.1 | 10.8 | 11.1 | 11.6 | 12.1 | 12.6 | 12.8 | 13.3 | 14.1 | 14.6 | 15.8 | 16.5 | 17.1 | 17.7 | 18.3 | 19.4 | 19.9 | 20.4 |
| 3.00 | 6.00 | 2.02 | 5.2 | 6.0 | 6.8 | 7.3 | 7.8 | 8.5 | 9.1 | 9.6 | 10.3 | 10.6 | 11.1 | 11.6 | 12.1 | 12.3 | 12.8 | 13.6 | 14.1 | 15.4 | 16.1 | 16.6 | 17.2 | 17.9 | 18.9 | 19.4 | 19.9 |
| * 2.35 | 4.75 | 2.04 | 6.8 | 7.6 | 8.3 | 8.8 | 9.3 | 10.1 | 10.7 | 11.1 | 11.9 | 12.1 | 12.7 | 13.1 | 13.6 | 13.9 | 14.4 | 15.1 | 15.6 | 16.9 | 17.6 | 18.1 | 18.7 | 19.4 | 20.4 | 20.9 | 21.4 |
| 6.90 | 14.00 | 2.04 | | | | | | | | | | | | | | | | | | | | | | | | | |
| * 2.20 | 4.50 | 2.07 | 7.1 | 7.9 | 8.7 | 9.2 | 9.7 | 10.4 | 11.0 | 11.4 | 12.2 | 12.4 | 13.0 | 13.4 | 13.9 | 14.2 | 14.7 | 15.4 | 15.9 | 17.2 | 17.9 | 18.5 | 19.1 | 19.7 | 20.7 | 21.2 | 21.7 |
| 3.15 | 6.50 | 2.08 | | 5.4 | 6.2 | 6.7 | 7.2 | 8.0 | 8.6 | 9.0 | 9.8 | 10.0 | 10.6 | 11.0 | 11.5 | 11.8 | 12.3 | 13.1 | 13.6 | 14.8 | 15.5 | 16.1 | 16.7 | 17.3 | 18.3 | 18.8 | 19.3 |
| 3.35 | 6.90 | 2.08 | | | 5.7 | 6.2 | 6.7 | 7.5 | 8.1 | 8.5 | 9.3 | 9.5 | 10.1 | 10.6 | 11.1 | 11.3 | 11.8 | 12.6 | 13.1 | 14.3 | 15.0 | 15.6 | 16.2 | 16.9 | 17.9 | 18.4 | 18.9 |
| 2.65 | 5.60 | 2.13 | 5.8 | 6.6 | 7.4 | 7.9 | 8.4 | 9.2 | 9.7 | 10.2 | 10.9 | 11.2 | 11.7 | 12.2 | 12.7 | 12.9 | 13.4 | 14.2 | 14.7 | 16.0 | 16.7 | 17.2 | 17.8 | 18.5 | 19.5 | 20.0 | 20.5 |
| 5.00 | 10.60 | 2.13 | | | | | | | | | | | | | | | | | 8.5 | 9.8 | 10.6 | 11.1 | 11.8 | 12.4 | 13.5 | 14.0 | 14.5 |
| * 2.50 | 5.30 | 2.14 | 6.2 | 7.0 | 7.7 | 8.3 | 8.8 | 9.5 | 10.1 | 10.5 | 11.3 | 11.5 | 12.1 | 12.5 | 13.0 | 13.3 | 13.8 | 14.6 | 15.1 | 16.3 | 17.0 | 17.6 | 18.2 | 18.8 | 19.8 | 20.3 | 20.8 |
| * 2.35 | 5.00 | 2.15 | 6.6 | 7.4 | 8.1 | 8.6 | 9.1 | 9.9 | 10.4 | 10.9 | 11.7 | 11.9 | 12.5 | 12.9 | 13.4 | 13.7 | 14.2 | 14.9 | 15.4 | 16.7 | 17.4 | 17.9 | 18.5 | 19.2 | 20.2 | 20.7 | 21.2 |
| 2.80 | 6.00 | 2.16 | 5.3 | 6.1 | 6.9 | 7.4 | 7.9 | 8.7 | 9.2 | 9.7 | 10.5 | 10.7 | 11.3 | 11.7 | 12.2 | 12.5 | 13.0 | 13.7 | 14.2 | 15.5 | 16.2 | 16.8 | 17.4 | 18.0 | 19.0 | 19.5 | 20.0 |
| 6.50 | 14.00 | 2.16 | | | | | | | | | | | | | | | | | | | | | | | | | |
| * 2.20 | 4.75 | 2.19 | 6.9 | 7.7 | 8.4 | 9.0 | 9.5 | 10.2 | 10.8 | 11.2 | 12.0 | 12.2 | 12.8 | 13.2 | 13.7 | 14.0 | 14.5 | 15.2 | 15.7 | 17.0 | 17.7 | 18.2 | 18.8 | 19.5 | 20.5 | 21.0 | 21.5 |
| 3.00 | 6.50 | 2.19 | | 5.5 | 6.3 | 6.8 | 7.3 | 8.1 | 8.7 | 9.1 | 9.9 | 10.1 | 10.7 | 11.2 | 11.7 | 11.9 | 12.4 | 13.2 | 13.7 | 14.9 | 15.6 | 16.2 | 16.8 | 17.5 | 18.5 | 19.0 | 19.5 |
| 3.15 | 6.90 | 2.21 | | | 5.8 | 6.3 | 6.8 | 7.6 | 8.2 | 8.7 | 9.4 | 9.7 | 10.2 | 10.7 | 11.2 | 11.5 | 12.0 | 12.7 | 13.2 | 14.5 | 15.2 | 15.7 | 16.3 | 17.0 | 18.0 | 18.5 | 19.0 |
| 3.65 | 8.00 | 2.21 | | | | | | | 6.8 | 7.3 | 8.1 | 8.3 | 8.9 | 9.3 | 9.9 | 10.1 | 10.6 | 11.4 | 11.9 | 13.2 | 13.9 | 14.4 | 15.0 | 15.7 | 16.7 | 17.2 | 17.7 |
| 4.75 | 10.60 | 2.24 | | | | | | | | | | | | | | | | 8.2 | 8.7 | 10.0 | 10.7 | 11.3 | 11.9 | 12.6 | 13.6 | 14.1 | 14.7 |
| * 2.50 | 5.60 | 2.27 | 5.9 | 6.7 | 7.5 | 8.0 | 8.5 | 9.3 | 9.8 | 10.3 | 11.0 | 11.3 | 11.8 | 12.3 | 12.8 | 13.0 | 13.5 | 14.3 | 14.8 | 16.1 | 16.8 | 17.3 | 17.9 | 18.6 | 19.6 | 20.1 | 20.6 |
| * 2.35 | 5.30 | 2.28 | 6.1 | 7.1 | 7.9 | 8.4 | 8.9 | 9.6 | 10.2 | 10.6 | 11.4 | 11.6 | 12.2 | 12.7 | 13.2 | 13.4 | 13.9 | 14.7 | 15.2 | 16.4 | 17.1 | 17.7 | 18.3 | 18.9 | 19.9 | 20.4 | 20.9 |
| 2.65 | 6.00 | 2.29 | 5.4 | 6.2 | 7.0 | 7.5 | 8.0 | 8.8 | 9.4 | 9.8 | 10.6 | 10.8 | 11.4 | 11.8 | 12.3 | 12.6 | 13.1 | 13.9 | 14.4 | 15.6 | 16.3 | 16.9 | 17.5 | 18.1 | 19.1 | 19.6 | 20.1 |
| * 2.20 | 5.00 | 2.30 | 6.7 | 7.5 | 8.2 | 8.7 | 9.2 | 10.0 | 10.6 | 11.0 | 11.8 | 12.0 | 12.6 | 13.0 | 13.5 | 13.8 | 14.3 | 15.0 | 15.5 | 16.8 | 17.5 | 18.0 | 18.6 | 19.3 | 20.3 | 20.8 | 21.3 |
| 3.00 | 6.90 | 2.32 | | | 5.9 | 6.4 | 6.9 | 7.7 | 8.3 | 8.8 | 9.5 | 9.8 | 10.3 | 10.8 | 11.3 | 11.6 | 12.1 | 12.8 | 13.3 | 14.6 | 15.3 | 15.9 | 16.5 | 17.1 | 18.1 | 18.6 | 19.1 |
| 6.00 | 14.00 | 2.34 | | | | | | | | | | | | | | | | | | | | | | | | | |
| 2.80 | 6.50 | 2.35 | | 5.6 | 6.4 | 6.9 | 7.5 | 8.2 | 8.8 | 9.3 | 10.0 | 10.3 | 10.8 | 11.3 | 11.8 | 12.1 | 12.6 | 13.3 | 13.8 | 15.1 | 15.8 | 16.3 | 16.9 | 17.6 | 18.6 | 19.1 | 19.6 |
| 10.60 | 25.00 | 2.36 | | | | | | | | | | | | | | | | | | | | | | | | | |
| 4.50 | 10.60 | 2.37 | | | | | | | | | | | | | | | | 8.3 | 8.9 | 10.2 | 10.9 | 11.5 | 12.1 | 12.8 | 13.8 | 14.3 | 14.8 |
| 8.00 | 19.00 | 2.38 | | | | | | | | | | | | | | | | | | | | | | | | | |
| * 2.35 | 5.60 | 2.41 | 6.0 | 6.8 | 7.6 | 8.1 | 8.6 | 9.4 | 9.9 | 10.4 | 11.1 | 11.4 | 11.9 | 12.4 | 12.9 | 13.2 | 13.7 | 14.4 | 14.9 | 16.2 | 16.9 | 17.4 | 18.0 | 18.7 | 19.7 | 20.2 | 20.7 |
| 3.35 | 8.00 | 2.41 | | | | | | 6.4 | 7.0 | 7.5 | 8.3 | 8.5 | 9.1 | 9.6 | 10.1 | 10.3 | 10.8 | 11.6 | 12.1 | 13.4 | 14.1 | 14.7 | 15.3 | 15.9 | 16.9 | 17.4 | 17.9 |
| * 2.50 | 6.00 | 2.43 | 5.5 | 6.3 | 7.1 | 7.6 | 8.1 | 8.9 | 9.5 | 9.9 | 10.7 | 10.9 | 11.5 | 11.9 | 12.5 | 12.7 | 13.2 | 14.0 | 14.5 | 15.7 | 16.4 | 17.0 | 17.6 | 18.2 | 19.2 | 19.7 | 20.2 |
| * 2.20 | 5.30 | 2.44 | 6.4 | 7.2 | 8.0 | 8.5 | 9.0 | 9.7 | 10.3 | 10.7 | 11.5 | 11.8 | 12.3 | 12.8 | 13.3 | 13.5 | 14.0 | 14.8 | 15.3 | 16.5 | 17.2 | 17.8 | 18.4 | 19.0 | 20.0 | 20.6 | 21.1 |
| 2.65 | 6.50 | 2.48 | | 5.7 | 6.5 | 7.0 | 7.6 | 8.3 | 8.9 | 9.4 | 10.1 | 10.4 | 10.9 | 11.4 | 11.9 | 12.2 | 12.7 | 13.4 | 13.9 | 15.2 | 15.9 | 16.5 | 17.1 | 17.7 | 18.7 | 19.2 | 19.7 |
| 2.80 | 6.90 | 2.49 | | | 6.0 | 6.6 | 7.1 | 7.9 | 8.4 | 8.9 | 9.7 | 9.9 | 10.5 | 10.9 | 11.4 | 11.7 | 12.2 | 13.0 | 13.5 | 14.7 | 15.4 | 16.0 | 16.6 | 17.3 | 18.3 | 18.8 | 19.3 |
| 5.60 | 14.00 | 2.51 | | | | | | | | | | | | | | | | | | | | | | | | | 19.8 |
| 3.15 | 8.00 | 2.56 | | | | | | 6.5 | 7.1 | 7.6 | 8.4 | 8.7 | 9.2 | 9.7 | 10.2 | 10.5 | 11.0 | 11.7 | 12.3 | 13.5 | 14.2 | 14.8 | 15.4 | 16.1 | 17.1 | 17.6 | 18.1 |
| * 2.20 | 5.60 | 2.58 | 6.1 | 6.9 | 7.7 | 8.2 | 8.7 | 9.5 | 10.0 | 10.5 | 11.2 | 11.5 | 12.1 | 12.5 | 13.0 | 13.3 | 13.8 | 14.5 | 15.0 | 16.3 | 17.0 | 17.5 | 18.1 | 18.8 | 19.8 | 20.3 | 20.8 |
| * 2.35 | 6.00 | 2.59 | 5.6 | 6.4 | 7.2 | 7.7 | 8.2 | 9.0 | 9.6 | 10.0 | 10.8 | 11.0 | 11.6 | 12.1 | 12.6 | 12.8 | 13.3 | 14.1 | 14.6 | 15.8 | 16.5 | 17.1 | 17.7 | 18.4 | 19.4 | 19.9 | 20.4 |
| 4.12 | 10.60 | 2.59 | | | | | | | | | | | | | | | | | 8.6 | 9.1 | 10.4 | 11.2 | 11.7 | 12.4 | 13.0 | 14.1 | 14.6 | 15.1 |
| * 2.50 | 6.50 | 2.63 | 5.0 | 5.8 | 6.6 | 7.1 | 7.7 | 8.4 | 9.0 | 9.5 | 10.2 | 10.5 | 11.0 | 11.5 | 12.0 | 12.3 | 12.8 | 13.5 | 14.0 | 15.3 | 16.0 | 16.6 | 17.2 | 17.8 | 18.8 | 19.3 | 19.8 |
| 2.65 | 6.90 | 2.63 | | 5.3 | 6.1 | 6.7 | 7.2 | 8.0 | 8.5 | 9.0 | 9.8 | 10.0 | 10.6 | 11.0 | 11.6 | 11.8 | 12.3 | 13.1 | 13.6 | 14.8 | 15.6 | 16.1 | 16.7 | 17.4 | 18.4 | 18.9 | 19.4 |
| 5.30 | 14.00 | 2.66 | | | | | | | | | | | | | | | | | | | | | | | | | 18.8 |
| 3.00 | 8.00 | 2.69 | | | | | | 6.6 | 7.2 | 7.7 | 8.5 | 8.8 | 9.3 | 9.8 | 10.3 | 10.6 | 11.1 | 11.8 | 12.4 | 13.6 | 14.3 | 14.9 | 15.5 | 16.2 | 17.2 | 17.7 | 18.2 |
| * 2.20 | 6.00 | 2.77 | 5.7 | 6.5 | 7.3 | 7.8 | 8.3 | 9.1 | 9.7 | 10.1 | 10.9 | 11.1 | 11.7 | 12.2 | 12.7 | 12.9 | 13.4 | 14.2 | 14.7 | 15.9 | 16.7 | 17.2 | 17.8 | 18.5 | 19.5 | 20.0 | 20.5 |
| 6.90 | 19.00 | 2.77 | | | | | | | | | | | | | | | | | | | | | | | | | |
| * 2.35 | 6.50 | 2.80 | 5.1 | 5.9 | 6.7 | 7.3 | 7.8 | 8.5 | 9.1 | 9.6 | 10.3 | 10.6 | 11.2 | 11.6 | 12.1 | 12.4 | 12.9 | 13.6 | 14.1 | 15.4 | 16.1 | 16.7 | 17.3 | 17.9 | 18.9 | 19.4 | 19.9 |
| * 2.50 | 6.90 | 2.80 | | 5.4 | 6.2 | 6.8 | 7.3 | 8.1 | 8.6 | 9.1 | 9.9 | 10.1 | 10.7 | 11.1 | 11.7 | 11.9 | 12.4 | 13.2 | 13.7 | 15.0 | 15.7 | 16.2 | 16.8 | 17.5 | 18.5 | 19.0 | 19.5 |
| 5.00 | 14.00 | 2.82 | | | | | | | | | | | | | | | | | | | | | | | | 19.6 | 19.2 |
| 2.80 | 8.00 | 2.89 | | | | | 5.9 | 6.8 | 7.4 | 7.8 | 8.6 | 8.9 | 9.5 | 9.9 | 10.4 | 10.7 | 11.2 | 12.0 | 12.5 | 13.8 | 14.5 | 15.0 | 15.7 | 16.3 | 17.3 | 17.8 | 18.3 |
| 3.65 | 10.60 | 2.93 | | | | | | | | | | | | | | | 8.0 | 8.9 | 9.4 | 10.7 | 11.5 | 12.1 | 12.7 | 13.4 | 14.4 | 14.9 | 15.4 |
| 6.50 | 19.00 | 2.94 | | | | | | | | | | | | | | | | | | | | | | | | | |
| 4.75 | 14.00 | 2.97 | | | | | | | | | | | | | | | | | | | | | | | 19.2 | 19.8 | 19.3 |
| * 2.35 | 6.90 | 2.98 | | 5.5 | 6.3 | 6.9 | 7.4 | 8.2 | 8.7 | 9.2 | 10.0 | 10.2 | 10.8 | 11.3 | 11.8 | 12.0 | 12.5 | 13.3 | 13.8 | 15.1 | 15.8 | 16.3 | 16.9 | 17.6 | 18.6 | 19.1 | 19.6 |
| * 2.20 | 6.50 | 3.00 | 5.2 | 6.0 | 6.8 | 7.4 | 7.9 | 8.6 | 9.2 | 9.7 | 10.4 | 10.7 | 11.3 | 11.7 | 12.2 | 12.5 | 13.0 | 13.7 | 14.3 | 15.5 | 16.2 | 16.8 | 17.4 | 18.0 | 19.0 | 19.5 | 20.1 |
| 2.65 | 8.00 | 3.06 | | | | | 6.0 | 6.9 | 7.4 | 7.9 | 8.7 | 9.0 | 9.6 | 10.0 | 10.5 | 10.8 | 11.3 | 12.1 | 12.6 | 13.9 | 14.6 | 15.1 | 15.8 | 16.4 | 17.4 | 17.9 | 18.4 |
| 4.50 | 14.00 | 3.13 | | | | | | | | | | | | | | | | | | | | | | | 19.4 | 19.9 | 19.5 |
| 8.00 | 25.00 | 3.14 | | | | | | | | | | | | | | | | | | | | | | | | | |
| 10.60 | 33.50 | 3.17 | | | | | | | | | | | | | | | | | | | | | | | | | |
| 6.00 | 19.00 | 3.18 | | | | | | | | | | | | | | | | | | | | | | | | | |
| * 2.20 | 6.90 | 3.19 | | 5.6 | 6.4 | 7.0 | 7.5 | 8.3 | 8.8 | 9.3 | 10.1 | 10.3 | 10.9 | 11.4 | 11.9 | 12.1 | 12.6 | 13.4 | 13.9 | 15.2 | 15.9 | 16.4 | 17.0 | 17.7 | 18.7 | 19.2 | 19.7 |
| 3.35 | 10.60 | 3.20 | | | | | | | | | | | | | | 7.7 | 8.2 | 9.1 | 9.6 | 10.9 | 11.7 | 12.3 | 12.9 | 13.6 | 14.6 | 15.1 | 15.6 |
| * 2.50 | 8.00 | 3.24 | | | | | 6.1 | 7.0 | 7.5 | 8.0 | 8.8 | 9.1 | 9.7 | 10.1 | 10.6 | 10.9 | 11.4 | 12.2 | 12.7 | 14.0 | 14.7 | 15.3 | 15.9 | 16.5 | 17.5 | 18.0 | 18.5 |
| 3.15 | 10.60 | 3.40 | | | | | | | | | | | | | 7.5 | 7.8 | 8.4 | 9.2 | 9.7 | 11.1 | 11.8 | 12.4 | 13.0 | 13.7 | 14.7 | 15.2 | 15.8 |
| 5.60 | 19.00 | 3.41 | | | | | | | | | | | | | | | | | | | | | | | | | |
| 4.12 | 14.00 | 3.43 | | | | | | | | | | | | | | | | | | | | | | | 19.6 | 19.4 | 19.1 |
| * 2.35 | 8.00 | 3.46 | | | | | 6.2 | 7.0 | 7.6 | 8.1 | 8.9 | 9.2 | 9.8 | 10.2 | 10.7 | 11.0 | 11.5 | 12.3 | 12.8 | 14.1 | 14.8 | 15.4 | 16.0 | 16.6 | 17.6 | 18.2 | 18.7 |

Key to Horsepower Correction Factor: 0.7 | 0.8 | 0.9 | 1.0 | 1.1 | 1.2

Includes both Wrap and Belt Length Correction Factors

## Table No. B6

# Super HC® V-Belt, Super HC Molded Notch® V-Belt, Predator® V-Belt, Super HC PowerBand® Belt, Super HC Molded Notch PowerBand Belt and Predator PowerBand Belt Drives

**Example 3VX800 Belt**
**Driver 6" sheave @1750rpm VR ~ 3.0**
**Rated Motor Power 2HP Light Shock 8hr/day**




From below:
d=6.0"
D = 19.0"
VR = 3.17

Fan Speed = 1750/3.17
Fan Speed = 552 rpm

$K_s$ = 1.2 (light shock)

$H_D$ = 1.1(2.0) = 2.2HP

Safety factor $n_{fs}$= 7.14/2.2
$n_{fs}$ = 3.25

V=πdn/12 = π(6)(1750)/12
V = 2750fpm < 5000 OK

Center Distance C = 19.3"

$C_\theta C_L$ = 0.9

V-Belt No. and Center Distance

| 3VX 550 | 3V 3VX 3VP 560 | 3VX 570 | 3VX 580 | 3VX 590 | 3V 3VX 3VP 600 | 3VX 616 | 3V 3VX 3VP 630 | 3V 3VX 650 | 3V 3VX 3VP 670 | 3VX 690 | 3V 3VX 3VP 710 | 3V 730 | 3V 3VX 3VP 750 | 3VX 771 | 3V 3VX 3VP 800 | 3V 810 | 3VX 826 | 3V 830 | 3V 3VX 3VP 850 | 3V 3VX 3VP 900 | 3VX 926 | 3V 3VX 3VP 950 | 3VX 974 | 3V 3VX 3VP 1000 | Speed Ratio | Sheave Outside Diameters: Small Sheave | Large Sheave |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 21.8 | 22.3 | 22.8 | 23.3 | 23.8 | 24.3 | 25.1 | 25.8 | 26.8 | 27.8 | 28.8 | 29.8 | | 31.8 | 32.8 | 34.3 | | 35.6 | | 36.8 | 39.3 | 40.6 | 41.8 | 43.0 | 44.3 | 1.92 | * 2.50 | 4.75 |
| 20.3 | 20.8 | 21.3 | 21.8 | 22.3 | 22.8 | 23.6 | 24.3 | 25.3 | 26.3 | 27.3 | 28.3 | 29.3 | 30.3 | 31.3 | 32.8 | 33.3 | 34.1 | 34.3 | 35.3 | 37.8 | 39.1 | 40.3 | 41.5 | 42.8 | 1.92 | 3.15 | 6.00 |
| 22.1 | 22.6 | 23.1 | 23.6 | 24.1 | 24.6 | 25.4 | 26.1 | 27.1 | 28.1 | 29.1 | 30.1 | | 32.1 | 33.2 | 34.6 | | 35.9 | | 37.1 | 39.6 | 40.9 | 42.1 | 43.3 | 44.6 | 1.93 | * 2.35 | 4.50 |
| 19.7 | 20.2 | 20.7 | 21.2 | 21.7 | 22.2 | 23.0 | 23.7 | 24.7 | 25.7 | 26.7 | 27.7 | 28.7 | 29.7 | 30.8 | 32.2 | 32.7 | 33.5 | 33.7 | 34.7 | 37.2 | 38.5 | 39.7 | 40.9 | 42.2 | 1.95 | 3.35 | 6.50 |
| 17.9 | 18.4 | 18.9 | 19.4 | 19.9 | 20.4 | 21.2 | 21.9 | 22.9 | 23.9 | 24.9 | 25.9 | 26.9 | 27.9 | 29.0 | 30.4 | 30.9 | 31.7 | 31.9 | 32.9 | 35.4 | 36.7 | 37.9 | 39.1 | 40.4 | 1.95 | 4.12 | 8.00 |
| 14.8 | 15.3 | 15.8 | 16.3 | | 17.3 | 18.1 | 18.8 | 19.8 | 20.8 | 21.9 | 22.9 | 23.9 | 24.9 | 25.9 | 27.4 | 27.9 | 28.7 | 28.9 | 29.9 | 32.4 | 33.7 | 34.9 | 36.1 | 37.4 | 2.01 | 5.30 | 10.60 |
| 21.6 | | 22.1 | 22.6 | 23.1 | 23.6 | 24.1 | 24.9 | 25.6 | 26.6 | 27.6 | 28.6 | 29.6 | | 31.6 | 32.6 | 34.1 | | 35.4 | | 36.6 | 39.1 | 40.4 | 41.6 | 42.8 | 44.1 | 2.02 | * 2.50 | 5.00 |
| 21.2 | 21.7 | 22.2 | 22.7 | 23.2 | 23.7 | 24.5 | 25.2 | 26.2 | 27.2 | 28.2 | 29.2 | 30.2 | 31.2 | 32.3 | 33.7 | 34.2 | 35.0 | 35.2 | 36.2 | 38.7 | 40.0 | 41.2 | 42.4 | 43.7 | 2.02 | 2.65 | 5.30 |
| 20.9 | 21.4 | 21.9 | 22.4 | 22.9 | 23.4 | 24.2 | 24.9 | 25.9 | 26.9 | 27.9 | 28.9 | 29.9 | 30.9 | 31.9 | 33.4 | 33.9 | 34.7 | 34.9 | 35.9 | 38.4 | 39.7 | 40.9 | 42.1 | 43.4 | 2.02 | 2.80 | 5.60 |
| 20.4 | 20.9 | 21.4 | 21.9 | 22.4 | 22.9 | 23.7 | 24.4 | 25.4 | 26.4 | 27.4 | 28.4 | 29.4 | 30.4 | 31.4 | 32.9 | 33.4 | 34.2 | 34.4 | 35.4 | 37.9 | 39.2 | 40.4 | 41.6 | 42.9 | 2.02 | 3.00 | 6.00 |
| 21.9 | 22.4 | 22.9 | 23.4 | 23.9 | 24.4 | 25.2 | 25.9 | 26.9 | 27.9 | 28.9 | 29.9 | | 31.9 | 33.0 | 34.4 | | 35.7 | | 36.9 | 39.4 | 40.7 | 41.9 | 43.1 | 44.4 | 2.04 | * 2.35 | 4.75 |
| | | 11.5 | 12.1 | 12.6 | 13.1 | 13.9 | 14.7 | 15.7 | 16.7 | 17.7 | 18.7 | 19.8 | 20.8 | 21.8 | 23.3 | 23.8 | 24.6 | 24.8 | 25.8 | 28.4 | 29.7 | 30.9 | 32.1 | 33.4 | 2.04 | 6.90 | 14.00 |
| 22.2 | 22.7 | 23.2 | 23.7 | 24.2 | 24.7 | 25.5 | 26.2 | 27.2 | 28.2 | 29.2 | 30.2 | | 32.2 | 33.3 | 34.7 | | 36.0 | | 37.2 | 39.7 | 41.0 | 42.2 | 43.4 | 44.7 | 2.07 | * 2.20 | 4.50 |
| 19.9 | 20.4 | 20.9 | 21.4 | 21.9 | 22.4 | 23.2 | 23.9 | 24.9 | 25.9 | 26.9 | 27.9 | 28.9 | 29.9 | 30.9 | 32.4 | 32.9 | 33.7 | 33.9 | 34.9 | 37.4 | 38.7 | 39.9 | 41.1 | 42.4 | 2.08 | 3.15 | 6.50 |
| 19.4 | 19.9 | 20.4 | 20.9 | 21.4 | 21.9 | 22.7 | 23.4 | 24.4 | 25.4 | 26.4 | 27.4 | 28.4 | 29.4 | 30.4 | 31.9 | 32.4 | 33.2 | 33.4 | 34.4 | 36.9 | 38.2 | 39.4 | 40.6 | 41.9 | 2.08 | 3.35 | 6.90 |
| 21.0 | 21.5 | 22.0 | 22.5 | 23.0 | 23.5 | 24.3 | 25.0 | 26.0 | 27.0 | 28.0 | 29.0 | 30.0 | 31.0 | 32.0 | 33.5 | 34.0 | 34.8 | 35.0 | 36.0 | 38.5 | 39.8 | 41.0 | 42.2 | 43.5 | 2.13 | 2.65 | 5.60 |
| 15.0 | 15.5 | 16.0 | 16.5 | 17.0 | 17.5 | 18.3 | 19.0 | 20.1 | 21.1 | 22.1 | 23.1 | 24.1 | 25.1 | 26.1 | 27.6 | 28.1 | 28.9 | 29.1 | 30.1 | 32.6 | 33.9 | 35.1 | 36.3 | 37.6 | 2.13 | 5.00 | 10.60 |
| 21.3 | 21.8 | 22.3 | 22.8 | 23.3 | 23.8 | 24.6 | 25.3 | 26.3 | 27.3 | 28.3 | 29.3 | | 31.3 | 32.4 | 33.8 | | 35.1 | | 36.3 | 38.8 | 40.1 | 41.4 | 42.6 | 43.9 | 2.14 | * 2.50 | 5.30 |
| 21.7 | 22.2 | 22.7 | 23.2 | 23.7 | 24.2 | 25.0 | 25.7 | 26.7 | 27.7 | 28.7 | 29.7 | | 31.7 | 32.8 | 34.2 | | 35.5 | | 36.7 | 39.2 | 40.5 | 41.7 | 42.9 | 44.2 | 2.15 | * 2.35 | 5.00 |
| 20.5 | 21.0 | 21.5 | 22.0 | 22.5 | 23.0 | 23.8 | 24.5 | 25.5 | 26.5 | 27.5 | 28.5 | 29.5 | 30.5 | 31.6 | 33.0 | 33.6 | 34.4 | 34.6 | 35.6 | 38.1 | 39.4 | 40.6 | 41.8 | 43.1 | 2.16 | 2.80 | 6.00 |
| | 11.3 | 11.8 | 12.3 | 12.8 | 13.4 | 14.2 | 14.9 | 16.0 | 17.0 | 18.0 | 19.0 | 20.0 | 21.1 | 22.1 | 23.6 | 24.1 | 24.9 | 25.1 | 26.1 | 28.7 | 30.0 | 31.2 | 32.4 | 33.7 | 2.16 | 6.50 | 14.00 |
| 22.0 | 22.5 | 23.0 | 23.5 | 24.0 | 24.5 | 25.3 | 26.0 | 27.0 | 28.0 | 29.0 | 30.0 | | 32.0 | 33.1 | 34.5 | | 35.8 | | 37.0 | 39.5 | 40.8 | 42.0 | 43.2 | 44.5 | 2.19 | * 2.20 | 4.75 |
| 20.0 | 20.5 | 21.0 | 21.5 | 22.0 | 22.5 | 23.3 | 24.0 | 25.0 | 26.0 | 27.0 | 28.0 | 29.0 | 30.0 | 31.0 | 32.5 | 33.0 | 33.8 | 34.0 | 35.0 | 37.5 | 38.8 | 40.0 | 41.2 | 42.5 | 2.19 | 3.00 | 6.50 |
| 19.5 | 20.0 | 20.5 | 21.0 | 21.5 | 22.0 | 22.8 | 23.5 | 24.5 | 25.5 | 26.5 | 27.5 | 28.5 | 29.5 | 30.6 | 32.1 | 32.6 | 33.4 | 33.6 | 34.6 | 37.1 | 38.4 | 39.6 | 40.8 | 42.1 | 2.21 | 3.15 | 6.90 |
| 18.2 | 18.7 | 19.2 | 19.7 | 20.2 | 20.7 | 21.5 | 22.2 | 23.2 | 24.3 | 25.3 | 26.3 | 27.3 | 28.3 | 29.3 | 30.8 | 31.3 | 32.1 | 32.3 | 33.3 | 35.8 | 37.1 | 38.3 | 39.5 | 40.8 | 2.21 | 3.65 | 8.00 |
| 15.2 | 15.7 | 16.2 | 16.7 | 17.2 | 17.7 | 18.5 | 19.2 | 20.2 | 21.2 | 22.3 | 23.3 | 24.3 | 25.3 | 26.3 | 27.8 | 28.3 | 29.1 | 29.3 | 30.3 | 32.8 | 34.1 | 35.3 | 36.5 | 37.8 | 2.24 | 4.75 | 10.60 |
| 21.1 | 21.6 | 22.1 | 22.6 | 23.1 | 23.6 | 24.4 | 25.1 | 26.1 | 27.1 | 28.1 | 29.1 | | 31.1 | 32.2 | 33.6 | | 34.9 | | 36.1 | 38.6 | 39.9 | 41.1 | 42.3 | 43.6 | 2.27 | * 2.50 | 5.60 |
| 21.4 | 21.9 | 22.4 | 22.9 | 23.4 | 23.9 | 24.7 | 25.4 | 26.5 | 27.5 | 28.5 | 29.5 | | 31.5 | 32.5 | 34.0 | | 35.3 | | 36.5 | 39.0 | 40.3 | 41.5 | 42.7 | 44.0 | 2.28 | * 2.35 | 5.30 |
| 20.6 | 21.1 | 21.6 | 22.1 | 22.6 | 23.1 | 23.9 | 24.6 | 25.7 | 26.7 | 27.7 | 28.7 | 29.7 | 30.7 | 31.7 | 33.2 | 33.7 | 34.5 | 34.7 | 35.7 | 38.2 | 39.5 | 40.7 | 41.9 | 43.2 | 2.29 | 2.65 | 6.00 |
| 21.8 | 22.3 | 22.8 | 23.3 | 23.8 | 24.3 | 25.1 | 25.8 | 26.8 | 27.8 | 28.8 | 29.8 | | 31.8 | 32.9 | 34.3 | | 35.6 | | 36.8 | 39.3 | 40.6 | 41.8 | 43.0 | 44.3 | 2.30 | * 2.20 | 5.00 |
| 19.6 | 20.1 | 20.6 | 21.1 | 21.6 | 22.1 | 22.9 | 23.6 | 24.6 | 25.7 | 26.7 | 27.7 | 28.7 | 29.7 | 30.7 | 32.2 | 32.7 | 33.5 | 33.7 | 34.7 | 37.2 | 38.5 | 39.7 | 40.9 | 42.2 | 2.32 | 3.00 | 6.90 |
| 11.1 | 11.6 | 12.1 | 12.7 | 13.2 | 13.7 | 14.5 | 15.3 | 16.3 | 17.3 | 18.4 | 19.4 | 20.4 | 21.4 | 22.5 | 24.0 | 24.5 | 25.3 | 25.5 | 26.5 | 29.0 | 30.3 | 31.5 | 32.7 | 34.1 | 2.34 | 6.00 | 14.00 |
| 20.1 | 20.6 | 21.1 | 21.6 | 22.1 | 22.6 | 23.4 | 24.1 | 25.1 | 26.1 | 27.1 | 28.1 | 29.1 | 30.1 | 31.2 | 32.6 | 33.1 | 33.9 | 34.1 | 35.1 | 37.7 | 39.0 | 40.2 | 41.4 | 42.7 | 2.35 | 2.80 | 6.50 |
| | | | | | | | | | | | | | | | | | | | | | | | 19.4 | 20.8 | 2.36 | 10.60 | 25.00 |
| 15.3 | 15.8 | 16.4 | 16.9 | 17.4 | 17.9 | 18.7 | 19.4 | 20.4 | 21.4 | 22.4 | 23.4 | 24.5 | 25.5 | 26.5 | 28.0 | 28.5 | 29.3 | 29.5 | 30.5 | 33.0 | 34.3 | 35.5 | 36.7 | 38.0 | 2.37 | 4.50 | 10.60 |
| | | | | | | | | | | | | | 15.3 | 16.4 | 17.9 | 18.5 | 19.3 | 19.5 | 20.6 | 23.1 | 24.5 | 25.7 | 26.9 | 28.3 | 2.38 | 9.00 | 19.00 |
| 21.2 | 21.7 | 22.2 | 22.7 | 23.2 | 23.7 | 24.5 | 25.2 | 26.2 | 27.2 | 28.2 | 29.2 | | 31.2 | 32.3 | 33.7 | | 35.0 | | 36.2 | 38.7 | 40.0 | 41.2 | 42.4 | 43.7 | 2.41 | * 2.35 | 5.60 |
| 18.4 | 18.9 | 19.4 | 20.0 | 20.5 | 21.0 | 21.8 | 22.5 | 23.5 | 24.5 | 25.5 | 26.5 | 27.5 | 28.5 | 29.5 | 31.0 | 31.5 | 32.3 | 32.5 | 33.5 | 36.0 | 37.3 | 38.5 | 39.7 | 41.0 | 2.41 | 3.35 | 8.00 |
| 20.8 | 21.3 | 21.8 | 22.3 | 22.8 | 23.3 | 24.1 | 24.8 | 25.8 | 26.8 | 27.8 | 28.8 | | 30.8 | 31.8 | 33.3 | | 34.6 | | 35.8 | 38.3 | 39.6 | 40.8 | 42.0 | 43.3 | 2.43 | * 2.50 | 6.00 |
| 21.6 | 22.1 | 22.6 | 23.1 | 23.6 | 24.1 | 24.9 | 25.6 | 26.6 | 27.6 | 28.6 | 29.6 | | 31.6 | 32.6 | 34.1 | | 35.4 | | 36.6 | 39.1 | 40.4 | 41.6 | 42.8 | 44.1 | 2.44 | * 2.20 | 5.30 |
| 20.2 | 20.7 | 21.2 | 21.7 | 22.2 | 22.7 | 23.5 | 24.2 | 25.2 | 26.2 | 27.2 | 28.2 | 29.3 | 30.3 | 31.3 | 32.8 | 33.3 | 34.1 | 34.3 | 35.3 | 37.8 | 39.1 | 40.3 | 41.5 | 42.8 | 2.48 | 2.65 | 6.50 |
| 19.8 | 20.3 | 20.8 | 21.3 | 21.8 | 22.3 | 23.1 | 23.8 | 24.8 | 25.8 | 26.8 | 27.8 | 28.8 | 29.8 | 30.9 | 32.3 | 32.8 | 33.6 | 33.8 | 34.8 | 37.3 | 38.6 | 39.8 | 41.0 | 42.3 | 2.49 | 2.80 | 6.90 |
| 11.3 | 11.9 | 12.4 | 12.9 | 13.4 | 14.0 | 14.8 | 15.5 | 16.6 | 17.6 | 18.6 | 19.7 | 20.7 | 21.7 | 22.8 | 24.2 | 24.7 | 25.6 | 25.8 | 26.8 | 29.3 | 30.6 | 31.8 | 33.0 | 34.3 | 2.51 | 5.60 | 14.00 |
| 18.6 | 19.1 | 19.6 | 20.1 | 20.6 | 21.1 | 21.9 | 22.6 | 23.6 | 24.6 | 25.6 | 26.6 | 27.6 | 28.6 | 29.7 | 31.1 | 31.6 | 32.5 | 32.7 | 33.7 | 36.2 | 37.5 | 38.7 | 39.9 | 41.2 | 2.56 | 3.15 | 8.00 |
| 21.3 | 21.8 | 22.3 | 22.8 | 23.3 | 23.8 | 24.6 | 25.3 | 26.3 | 27.3 | 28.3 | 29.3 | | 31.3 | 32.4 | 33.8 | | 35.1 | | 36.3 | 38.8 | 40.1 | 41.3 | 42.5 | 43.8 | 2.58 | * 2.20 | 5.60 |
| 20.9 | 21.4 | 21.9 | 22.4 | 22.9 | 23.4 | 24.2 | 24.9 | 25.9 | 26.9 | 27.9 | 28.9 | | 30.9 | 31.9 | 33.4 | | 34.7 | | 35.9 | 38.4 | 39.7 | 40.9 | 42.1 | 43.4 | 2.59 | * 2.35 | 6.00 |
| 15.6 | 16.1 | 16.6 | 17.1 | 17.6 | 18.1 | 19.0 | 19.7 | 20.7 | 21.7 | 22.7 | 23.7 | 24.7 | 25.7 | 26.8 | 28.3 | 28.8 | 29.6 | 29.8 | 30.8 | 33.3 | 34.6 | 35.8 | 37.0 | 38.3 | 2.59 | 4.12 | 10.60 |
| 20.3 | 20.8 | 21.3 | 21.8 | 22.3 | 22.8 | 23.6 | 24.3 | 25.4 | 26.4 | 27.4 | 28.4 | | 30.4 | 31.4 | 32.9 | | 34.2 | | 35.4 | 37.9 | 39.2 | 40.4 | 41.6 | 42.9 | 2.63 | * 2.50 | 6.50 |
| 19.9 | 20.4 | 20.9 | 21.4 | 21.9 | 22.4 | 23.2 | 23.9 | 24.9 | 25.9 | 26.9 | 27.9 | 28.9 | 29.9 | 31.0 | 32.4 | 32.9 | 33.7 | 33.9 | 34.9 | 37.4 | 38.7 | 39.9 | 41.1 | 42.4 | 2.63 | 2.65 | 6.90 |
| 11.5 | 12.0 | 12.6 | 13.1 | 13.6 | 14.2 | 15.0 | 15.7 | 16.8 | 17.8 | 18.8 | 19.9 | 20.9 | 21.9 | 23.0 | 24.5 | 25.0 | 25.8 | 26.0 | 27.0 | 29.5 | 30.8 | 32.0 | 33.3 | 34.6 | 2.66 | 5.30 | 14.00 |
| 18.7 | 19.2 | 19.7 | 20.2 | 20.7 | 21.2 | 22.0 | 22.7 | 23.7 | 24.7 | 25.7 | 26.7 | 27.7 | 28.8 | 29.8 | 31.3 | 31.8 | 32.6 | 32.8 | 33.8 | 36.3 | 37.6 | 38.8 | 40.0 | 41.3 | 2.69 | 3.00 | 8.00 |
| 21.0 | 21.5 | 22.0 | 22.5 | 23.0 | 23.5 | 24.3 | 25.0 | 26.0 | 27.0 | 28.0 | 29.0 | | 31.0 | 32.1 | 33.5 | | 34.8 | | 36.0 | 38.5 | 39.8 | 41.0 | 42.2 | 43.5 | 2.77 | * 2.20 | 6.00 |
| | | | | | | | | | | | 13.8 | 14.9 | 16.0 | 17.1 | 18.7 | 19.2 | 20.0 | 20.2 | 21.3 | 23.9 | 25.2 | 26.5 | 27.7 | 29.0 | 2.77 | 6.90 | 19.00 |
| 20.4 | 20.9 | 21.4 | 22.0 | 22.5 | 23.0 | 23.8 | 24.5 | 25.5 | 26.5 | 27.5 | 28.5 | | 30.5 | 31.5 | 33.0 | | 34.3 | | 35.5 | 38.0 | 39.3 | 40.5 | 41.7 | 43.0 | 2.80 | * 2.35 | 6.50 |
| 20.0 | 20.5 | 21.0 | 21.5 | 22.0 | 22.5 | 23.3 | 24.0 | 25.0 | 26.0 | 27.0 | 28.0 | | 30.0 | 31.1 | 32.5 | | 33.8 | | 35.0 | 37.6 | 38.9 | 40.1 | 41.3 | 42.6 | 2.80 | * 2.50 | 6.90 |
| 11.7 | 12.2 | 12.8 | 13.3 | 13.8 | 14.4 | 15.2 | 15.9 | 17.0 | 18.0 | 19.0 | 20.1 | 21.1 | 22.1 | 23.2 | 24.7 | 25.2 | 26.0 | 26.2 | 27.2 | 29.7 | 31.1 | 32.3 | 33.5 | 34.8 | 2.82 | 5.00 | 14.00 |
| 18.8 | 19.3 | 19.8 | 20.4 | 20.9 | 21.4 | 22.2 | 22.9 | 23.9 | 24.9 | 25.9 | 26.9 | 27.9 | 28.9 | 30.0 | 31.4 | 31.9 | 32.7 | 32.9 | 33.9 | 36.4 | 37.7 | 38.9 | 40.1 | 41.4 | 2.89 | 2.80 | 8.00 |
| 15.9 | 16.4 | 17.0 | 17.5 | 18.0 | 18.5 | 19.3 | 20.0 | 21.0 | 22.0 | 23.0 | 24.1 | 25.1 | 26.1 | 27.1 | 28.6 | 29.1 | 29.9 | 30.1 | 31.1 | 33.6 | 34.9 | 36.1 | 37.3 | 38.7 | 2.93 | 3.65 | 10.60 |
| | | | | | | | | | | | 14.1 | 15.2 | 16.3 | 17.4 | 18.9 | 19.5 | 20.3 | 20.5 | 21.6 | 24.2 | 25.5 | 26.7 | 28.0 | 29.3 | 2.94 | 6.50 | 19.00 |
| 11.9 | 12.4 | 12.9 | 13.5 | 14.0 | 14.5 | 15.4 | 16.1 | 17.1 | 18.2 | 19.2 | 20.2 | 21.3 | 22.3 | 23.4 | 24.8 | 25.4 | 26.2 | 26.4 | 27.4 | 29.9 | 31.2 | 32.4 | 33.7 | 35.0 | 2.97 | 4.75 | 14.00 |
| 20.1 | 20.6 | 21.1 | 21.6 | 22.1 | 22.6 | 23.4 | 24.1 | 25.1 | 26.1 | 27.1 | 28.1 | | 30.1 | 31.2 | 32.7 | | 34.0 | | 35.2 | 37.7 | 39.0 | 40.2 | 41.4 | 42.7 | 2.98 | * 2.35 | 6.90 |
| 20.6 | 21.1 | 21.6 | 22.1 | 22.6 | 23.1 | 23.9 | 24.6 | 25.6 | 26.6 | 27.6 | 28.6 | | 30.6 | 31.6 | 33.1 | | 34.4 | | 35.6 | 38.1 | 39.4 | 40.6 | 41.8 | 43.1 | 3.00 | * 2.20 | 6.50 |
| 18.9 | 19.5 | 20.0 | 20.5 | 21.0 | 21.5 | 22.3 | 23.0 | 24.0 | 25.0 | 26.0 | 27.0 | 28.0 | 29.0 | 30.1 | 31.5 | 32.0 | 32.8 | 33.0 | 34.0 | 36.5 | 37.8 | 39.0 | 40.2 | 41.5 | 3.06 | 2.65 | 8.00 |
| 17.0 | 17.6 | 18.1 | 18.6 | 19.1 | 19.7 | 20.5 | 21.3 | 22.3 | 23.3 | 24.3 | 25.3 | 26.3 | 27.3 | 28.4 | 29.8 | 30.3 | 31.1 | 31.3 | 32.3 | 34.8 | 36.1 | 37.3 | 38.5 | 39.8 | 3.13 | 4.50 | 14.00 |
| | | | | | | | | | | | | | | | | | | | | | | 18.4 | 19.7 | 21.0 | 22.5 | 3.14 | 8.00 | 25.00 |
| | | | | | | | | | | | 14.4 | 15.5 | 16.6 | 17.7 | 19.2 | 19.8 | 20.6 | 20.8 | 21.9 | 24.5 | 25.8 | 27.1 | 28.3 | 29.6 | 3.16 | 6.00 | 19.00 |
| 20.2 | 20.7 | 21.2 | 21.7 | 22.2 | 22.7 | 23.5 | 24.2 | 25.2 | 26.2 | 27.3 | 28.3 | | 30.3 | 31.3 | 32.8 | | 34.1 | | 35.3 | 37.8 | 39.1 | 40.3 | 41.5 | 42.8 | 3.19 | * 2.20 | 6.90 |
| 16.1 | 16.6 | 17.2 | 17.7 | 18.2 | 18.7 | 19.5 | 20.2 | 21.2 | 22.2 | 23.3 | 24.3 | 25.3 | 26.3 | 27.4 | 28.8 | 29.3 | 30.1 | 30.3 | 31.3 | 33.8 | 35.2 | 36.4 | 37.6 | 38.9 | 3.20 | 3.35 | 10.60 |
| 19.1 | 19.6 | 20.1 | 20.6 | 21.1 | 21.6 | 22.4 | 23.1 | 24.1 | 25.1 | 26.1 | 27.1 | | 29.1 | 30.2 | 31.6 | | 32.9 | | 34.1 | 36.7 | 38.0 | 39.2 | 40.4 | 41.7 | 3.24 | * 2.50 | 8.00 |
| 16.3 | 16.8 | 17.3 | 17.8 | 18.3 | 18.8 | 19.6 | 20.4 | 21.4 | 22.4 | 23.4 | 24.4 | 25.4 | 26.4 | 27.5 | 29.0 | 29.5 | 30.3 | 30.5 | 31.5 | 34.0 | 35.3 | 36.5 | 37.7 | 39.0 | 3.40 | 3.15 | 10.60 |
| | | | | | | | | | | | 13.5 | 14.6 | 15.7 | 16.8 | 18.0 | 19.5 | 20.0 | 20.9 | 21.1 | 22.2 | 24.8 | 26.1 | 27.4 | 28.6 | 29.9 | 3.41 | 5.60 | 19.00 |
| 12.3 | 12.8 | 13.3 | 13.9 | 14.4 | 14.9 | 15.8 | 16.5 | 17.6 | 18.6 | 19.6 | 20.7 | 21.7 | 22.7 | 23.8 | 25.3 | 25.8 | 26.6 | 26.8 | 27.8 | 30.4 | 31.7 | 32.9 | 34.1 | 35.4 | 3.43 | 4.12 | 14.00 |
| 19.2 | 19.7 | 20.2 | 20.7 | 21.2 | 21.7 | 22.5 | 23.2 | 24.2 | 25.2 | 26.2 | 27.2 | | 29.2 | 30.3 | 31.7 | | 33.1 | | 34.3 | 36.8 | 38.1 | 39.3 | 40.5 | 41.8 | 3.46 | * 2.35 | 8.00 |

Key to Horsepower Correction Factor: 0.7 | 0.8 | 0.9 | 1.0 | 1.1 | 1.2

$P_A = C_\theta C_L (P_{Tab} + \Delta P_R) = 0.9(7.68+0.25)=7.14HP$

# Regular V-Belt Tensioning Method

## Step 1 Calculate the Required Base Static Installation Tension Per Strand of Belt (Static Tension)

**A.** The **static tension per strand** ($T_{st}$) is given by this formula:

### Formula No. D19

$$T_{st} = 15\left(\frac{2.5^{*} - K\phi}{K\phi}\right)\left(\frac{(\text{Motor HP})\ (10^3)}{(N)(V)}\right) + \frac{MV^2}{10^6}$$

Where: Kø = arc correction factor from Table No. D26 or Table No. D11 on Page D12 for V-Flat drives.

N = Number of belts. (This is the number of strands in the case of PowerBand® Belts.)

V = Belt speed, ft./min.

M = Constant from Table No. D27.

*2.67 for Micro-V® Belts.

### Table No. D26

**Arc of Contact Correction Factor Kø for V-V Drives**

| $\frac{D-d}{C}$ | Arc of Contact on Small Sheave (°) | Factor Kø: A, B, C, D 3V, 5V, 8V 5M, 7M, 11M | Factor Kø: Micro-V J, L, M |
|---|---|---|---|
| 0.00 | 180 | 1.00 | 1.00 |
| 0.10 | 174 | 0.99 | 0.98 |
| 0.20 | 169 | 0.97 | 0.97 |
| 0.30 | 163 | 0.96 | 0.95 |
| 0.40 | 157 | 0.94 | 0.94 |
| 0.50 | 151 | 0.93 | 0.92 |
| 0.60 | 145 | 0.91 | 0.90 |
| 0.70 | 139 | 0.89 | 0.88 |

(D-d)/C
=(19-6)/19.3
=0.67

So $K_\phi$ ~ 0.90

### Table No. D27

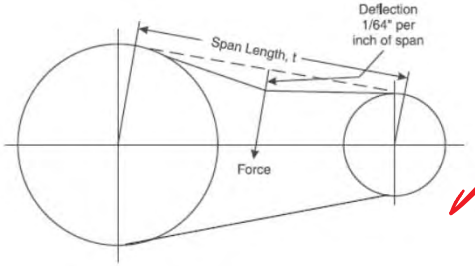
**Factor M and Factor Y**

| Cross Section | M | Y |
|---|---|---|
| Super HC® Molded Notch | | |
| 3VX | 0.29 | 4 |
| 5VX | 0.78 | 13 |
| Super HC® Molded Notch PowerBand® | | |
| 3VX | 0.39 | 4 |
| 5VX | 0.98 | 13 |
| Super HC | | |
| 5V | 1.0 | 11 |
| 8V | 2.6 | 22 |
| Super HC PowerBand | | |
| 3V | 0.46 | 4 |
| 5V | 1.2 | 11 |
| 5VP | 1.2 | 39 |
| 8V | 3.0 | 22 |

| Cross Section | M | Y |
|---|---|---|
| Tri-Power® Molded Notch | | |
| AX | 0.47 | 7 |
| BX | 0.76 | 8 |
| CX | 1.31 | 15 |
| Micro-V® Belt | | |
| J* | 0.035 | 0.56 |
| L | 0.130 | 1.90 |
| M | 0.520 | 6.30 |
| Polyflex® JB® | | |
| 5M** | 0.05 | 1.2 |
| 7M | 0.14 | 4.6 |
| 11M | 0.31 | 8.5 |
| Predator Singles | | |
| AP | 0.56 | 21 |

## Step 2 Calculate the Minimum and Maximum Recommended Forces to Deflect One Belt 1/64" Per Inch of Span Length

**A.** Measure the **span length** (t) of your drive (see sketch).




$$t = \sqrt{C^2 - \left(\frac{D-d}{2}\right)^2}$$

Formula incorrect

### Span Length, Two Wheel Drives

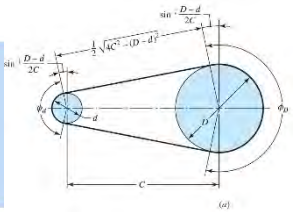
Belt span length is needed for the deflection method of measuring V-belt installation tension. Span length can be measured on the drive or measured from a scale layout of the drive.

For V or V-flat drives using only two wheels (no idlers) span length can be calculated from the following formula:

### Formula No. D35

$$t = C\left[1 - 0.125\left(\frac{D-d}{C}\right)^2\right]$$

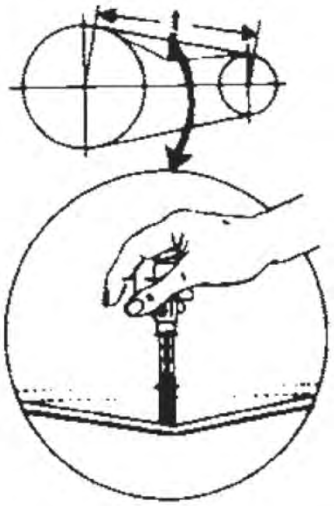
where: t = span length, inches
C = center distance, inches
D = large sheave or pulley diameter, inches
d = small sheave diameter, inches

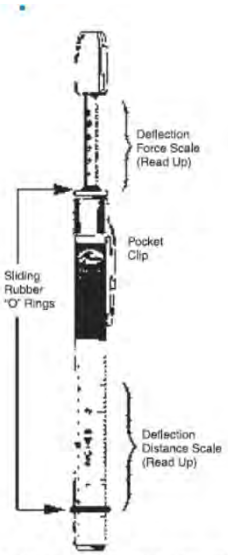


NOTE: D and d are Outside Diameters for Super HC and Datum Diameters for Hi-Power II and Tri-Power Molded Notch V-Belts.

D = 19.0", d = 6.0", C = 19.3" $K_\phi$ = 0.90, V= 2750fpm
N = 1 (one belt), Motor HP = $H_{nom}$ = 2 HP
M = 0.29 (Table D27 for 3VX Belt)

Span length t = 18.21" (Formula D35)
Static Tension $T_{st}$ = 21.58 lbs (Formula D19)

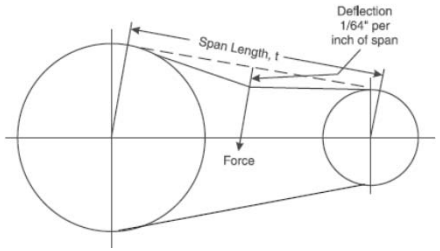





Read the scales at the bottom edge of the "O" Ring. Leave the upper "O" Ring in maximum "down" position

| Part No. | Force Limitati |
|---|---|
| 7401-0076 | Up to 30 lb |
| 7401-0075* | Up to 66 lb |

*Dual Tensioner




L= 80", t=18.21", $T_{st}$ = 21.58 lbs, Y = 4 (Table D27 for 3VX Belt)
One belt is used for this application.

Min Force = 1.95 lbs  Max. Force = 2.08 lbs  (Formula D22/23)

A force of  2lbs applied at the center of the span
to deflect belt approx. 18.21/64" (0.28" or 7.2mm.)..

## Step 2 Calculate the Minimum and Maximum Recommended Forces to Deflect One Belt 1/64" Per Inch of Span Length

**A.** Measure the **span length** (t) of your drive (see sketch).

**B.** If your drive uses **two or more** PowerBand Belts or individual belts, calculate the lower and upper recommended deflection forces by these formulas:

### Formula No. D20

$$\text{Minimum Recommended Force} = \frac{1.4\,T_{st} + Y}{16}$$

### Formula No. D21

$$\text{Maximum Recommended Force} = \frac{1.5\,T_{st} + Y}{16}$$

Where: $T_{st}$ = tension per strand from Step 1.
$Y$ = constant from Table No. D27.

**C.** If your drive has only **one** PowerBand Belt (See Step D) or individual belt, calculate the lower and upper recommended deflection forces by these formulas:

### Formula No. D22

$$\text{Minimum Recommended Force} = \frac{1.4\,T_{st} + \left(\frac{t}{L}\right)Y}{16}$$

### Formula No. D23

$$\text{Upper Recommended Force} = \frac{1.5\,T_{st} + \left(\frac{t}{L}\right)Y}{16}$$

Where: $T_{st}$ = tension per strand from Step 1.
$Y$ = constant from Table No. D27.
$t$ = span length (see Figure No. D25).
$L$ = belt length

**D.** The deflection forces calculated in Step 2B or 2C are for an individual belt. Multiply these forces by the number of individual strands in a band to get the lower and upper recommended forces for a PowerBand Belt. (If your drive uses 2 or more PowerBand Belts, use the band with the fewest number of strands.)

# V-Belt Design (Shigley)

$H_a = K_1 K_2 H_{tab}$

$H_{TAB} = P_{TAB}$ Design Power (Catalogue)

$H_a$ Allowable Power

$K_1$ wrap angle correction (Table 17-13)

$K_2$ belt length correction (Table 17-14)

$H_d = H_{nom} K_s n_d$

$H_{nom}$ = Nominal Power (motor rating)

$H_d$ = Design Power

$K_s$ service factor (next slide)

$n_d$ design ~~safety~~ factor

$N_b > H_d / H_a$

$N_b$ = required number of belts

Safety Factor $n_{fs} = N_b H_a / H_d$

# V-Belt Analysis Procedure

1. Find geometry and speed ($L_p$, C, φ, & V)
2. Find design and allowable power ($H_d$ & $H_a$)
3. Find required number of belts
4. Find belt tensions ($F_c$, $F_1$, $F_2$, and $F_i$)
5. Check if $f/\sin(\beta/2) > f'$ ($f'$ = friction development)
6. Find safety factor ($n_{fs}$)
7. Find belt life (see Example 17-4)